

Engineering, Procurement and Construction Contract for the Alpilla 600 MW Combined Cycle Power Plant

Contract No. ALP-EPC-001

between

Alpilla Enerji Üretim A.Ş.

(the Owner)

and

the Consortium of

Istanbul EPC Mühendislik ve Taahhüt A.Ş. (Leader)

Imaginary Electric Company

(jointly and severally, the Contractor)

Form	Lump-sum, fixed-price, date-certain turnkey EPC contract with a consortium Contractor (joint and several liability)
Contract Price	EUR 532,480,000 (lump sum, fixed)
Price basis	EUR 515,000,000 initial negotiated price (gas only); +EUR 16,000,000 LDO secondary fuel (Owner revision B) = EUR 531,000,000; +EUR 1,480,000 agreed clarifications (TQ-007 stack 65 m EUR 180,000; TQ-008 sanitary treatment EUR 350,000; TQ-018 LDO treatment cost share EUR 950,000) = EUR 532,480,000 (agreed 2026-09-27).
Date of signature	2026-10-15
Notice to Proceed	2026-11-02
Time for Completion	975 days from NTP
Governing law	Laws of the Republic of Türkiye
Status	for signature

CASE STUDY. All parties, names, prices and dates are fictional and created to exercise the project environment.

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Part I - Contract Agreement

Article A1 Parties

This Contract is made between Alpilla Enerji Üretim A.Ş., a company incorporated in the Republic of Türkiye (the "Owner"), and the unincorporated consortium formed by Istanbul EPC Mühendislik ve Taahhüt A.Ş. (the "Leader") and Imaginary Electric Company (the "Member"), acting jointly and severally (together the "Contractor").

This is a case-study document. All parties, prices and dates are fictional.

Article A2 Recitals

- The Owner intends to develop, own and operate a natural-gas-fired combined cycle power plant of 600 MW net output on the Site on the northern Marmara coast (the "Project").
- The Owner invited tenders for the engineering, procurement and construction of the Project on a lump-sum turnkey basis and accepted the Contractor's tender as clarified during negotiations.
- The Contractor has examined the Site, the Owner's Requirements and all information made available and is satisfied as to the sufficiency of the Contract Price.

Article A3 Contract documents

The following documents form the Contract and shall be read as one. In case of conflict they take precedence in the order listed:

- this Contract Agreement;
- the Conditions of Contract (Part II);
- Appendix I - Agreed Clarifications (prevailing over Appendix A and the Owner's documents where they modify them);
- Appendix B - Performance Guarantees and Liquidated Damages;
- Appendix C - Schedule of Payments;
- Appendix D - Scope of Works Split, Terminal Points and Tie-in Register;
- Appendix A - Employer's Requirements (technical);
- Appendix E - Site and Design Data;
- Appendix H - Owner-provided Documents;
- Appendix G - Key Dates;
- the Contractor's technical proposal, to the extent it is consistent with the documents above.

Article A4 Contract Price

In consideration of the Works the Owner shall pay the Contractor the lump-sum Contract Price of EUR 532,480,000 (five hundred and thirty-two million four hundred and eighty thousand euros), comprising an offshore portion of EUR 327,000,000 and an onshore portion of EUR 205,480,000, exclusive of Value Added Tax, payable in accordance with Clause 15 and Appendix C. The Contract Price is fixed and not subject to escalation or exchange-rate adjustment, except as adjusted under Clause 14.

Article A5 Time for Completion

The Contractor shall complete the Works, including passing the Tests on Completion and the Reliability Run, within 975 days from the Notice to Proceed (NTP), and shall achieve the Key Dates in Appendix G.

Article A6 Effectiveness

This Contract enters into force on signature. The Notice to Proceed shall be issued within 21 days after the last of the following conditions precedent has been fulfilled: (a) delivery of the Performance Security and the Advance Payment Guarantee; (b) delivery of the parent company guarantees of both consortium members; (c) financial close of the Owner's project financing; and (d) the positive EIA decision remaining valid and unchallenged.

Article A7 Signatures

In witness whereof the Parties have caused this Contract to be executed by their duly authorised representatives on the date stated on the cover page.

For the Owner Alpilla Enerji Üretim A.Ş.	For the Contractor (Leader) Istanbul EPC Mühendislik ve Taahhüt A.Ş.	For the Contractor (Member) Imaginary Electric Company
Name:	Name:	Name:
Title:	Title:	Title:
Signature:	Signature:	Signature:
Date:	Date:	Date:

Part II - Conditions of Contract

1 General Provisions

1.1 Definitions

In the Contract the following words have the meanings stated, unless the context requires otherwise:

- "Works" means the permanent works and temporary works required to deliver the complete power plant described in the Employer's Requirements.
- "Site" means the land made available by the Owner as described in ER-02.01, including the marine areas for intake and outfall.
- "Engineer" means the person appointed by the Owner to act as the Owner's engineer and notified to the Contractor.
- "Taking-Over" means the stage at which the Owner takes over the Works under Clause 11.
- "Defects Notification Period" means the period stated in Sub-Clause 13.1.
- "Site Reference Conditions" means the conditions stated in ER-02.03.
- "Good Industry Practice" means the skill, diligence and foresight reasonably expected of an experienced international contractor executing works of similar size and complexity.
- "Owner's Database" means the engineering database designated by the Owner as the single source of project data.

1.2 Interpretation

Words in the singular include the plural and vice versa. Headings do not affect interpretation. "Including" means including without limitation. A reference to a law, code or standard is to the version in force at the Base Date (thirty days before the tender submission date), unless stated otherwise.

1.3 Priority of documents

The documents forming the Contract are mutually explanatory. Ambiguities or discrepancies shall be resolved by the order of precedence in Article A3. Where a discrepancy exists within the Employer's Requirements, the more stringent requirement applies. The Contractor shall notify any discrepancy it discovers without delay.

1.4 Law and language

The Contract is governed by the laws of the Republic of Türkiye. The ruling language is English. Communications with Turkish authorities shall be in Turkish, prepared by the Contractor where the Contractor is responsible for the underlying permit or approval.

1.5 Communications

Notices, consents, approvals and instructions shall be in writing and delivered through the project document management system or by hand to the addresses in Appendix F. A notice is effective when received. Informal communications, including meeting minutes, do not constitute a notice unless expressly stated.

1.6 Confidentiality and data

Each Party shall keep the other Party's confidential information confidential and use it only for the Project. The Owner may use all design data, documents and database content delivered by the Contractor for the operation, maintenance, repair, extension and refinancing of the Project.

2 The Owner

2.1 Access to the Site

The Owner shall give the Contractor access to and possession of the Site, levelled to +15.0 m, within 14 days after NTP, and access to the marine areas once the relevant permits are obtained under Sub-Clause 2.2.

2.2 Permits

The Owner shall obtain and maintain the EIA decision, the generation licence, the grid connection agreement and the land-use permits. The Contractor shall obtain all other permits for the design, construction and testing of the Works and shall provide all technical inputs the Owner needs for the Owner's permits.

2.3 Owner's financial arrangements

The Owner shall, within 28 days after receiving a reasonable request, provide reasonable evidence that financial arrangements are in place to pay the Contract Price.

2.4 Owner-furnished items and utilities

The Owner shall provide the items allocated to it in Appendix D, including natural gas for commissioning and tests, grid access for back-feed and test energy, and utilities to the Site boundary, at the times required by the programme accepted under Sub-Clause 9.3.

3 The Engineer

3.1 Role of the Engineer

The Engineer acts for the Owner in reviewing documents, inspecting and witnessing tests, certifying payments and determining claims. The Engineer has no authority to amend the Contract. Any approval, review or inspection by the Engineer does not relieve the Contractor of any responsibility.

3.2 Instructions

The Contractor shall comply with instructions of the Engineer given in accordance with the Contract. If the Contractor considers that an instruction constitutes a Variation, it shall notify the Engineer within 14 days before commencing the work, unless the instruction concerns safety.

4 The Contractor

4.1 General obligations

The Contractor shall execute the Works in accordance with the Contract, Good Industry Practice and applicable laws, so that on completion the Works are fit for the purposes defined in the Employer's Requirements. The Contractor shall provide all design, plant, materials, labour, equipment and services required, whether temporary or permanent.

4.2 Performance Security

The Contractor shall deliver an unconditional, irrevocable on-demand bank guarantee from a bank acceptable to the Owner for 10 % of the Contract Price, valid until the issue of the Performance Certificate, reducing to 5 % at Taking-Over.

4.3 Contractor's representative

The Leader shall appoint a Project Director resident on Site with full authority to act for the Contractor. Replacement requires the Owner's prior consent, not to be unreasonably withheld.

4.4 Subcontractors and vendors

The Contractor shall not subcontract the whole of the Works. Subcontracting of the items listed as major equipment in ER-01.08 and of civil works above EUR 5,000,000 requires the Owner's prior consent. The Contractor remains responsible for all subcontractors.

4.5 Site data

The Owner has made available the Site data listed in Appendix E. The Contractor is responsible for interpreting that data and for any additional investigation. The Contractor is entitled to relief only for physical

conditions that an experienced contractor could not have foreseen at the Base Date after reasonable investigation.

4.6 Health, safety and environment

The Contractor shall comply with all HSE laws, the EIA conditions and the Owner's HSE requirements, and shall stop any unsafe work. Serious incidents shall be reported to the Owner within 24 hours.

4.7 Quality management

The Contractor shall implement the quality system required by ER-15 and give the Engineer access to all places of manufacture. Compliance with the quality system does not relieve the Contractor of any obligation.

4.8 Local content

The Contractor shall use reasonable endeavours to procure goods and services in Türkiye where competitive in quality, price and delivery, and shall report the local content achieved each quarter.

4.9 Engineering data

The Contractor shall deliver all engineering data into the Owner's Database in accordance with ER-01.06, ER-01.10 and ER-18.04. Data in the Owner's Database prevails over any conflicting copy.

5 The Consortium

5.1 Joint and several liability

The Leader and the Member are jointly and severally liable to the Owner for the performance of all obligations of the Contractor. The Owner may claim against either or both of them.

5.2 Authority of the Leader

The Leader is authorised to bind the Contractor, to receive notices and instructions and to receive payments for the Contractor. Notices given to the Leader are deemed given to both members.

5.3 Composition

The composition of the consortium and the allocation of work set out in Appendix D shall not be changed without the Owner's prior consent.

5.4 Internal arrangements

The consortium agreement between the Leader and the Member governs only their internal relationship. No provision of it limits their liability towards the Owner.

5.5 Insolvency of a member

If a member becomes insolvent or ceases to perform, the other member shall complete the Works without additional cost to the Owner, with a replacement acceptable to the Owner where required for the power island.

6 Design

6.1 General design obligations

The Contractor shall carry out and be responsible for the design of the Works, performed by qualified engineers. The design shall comply with the Employer's Requirements and the laws and codes stated therein.

6.2 Contractor's documents and review

The Contractor shall submit the documents listed in the master document register for review. The Engineer shall respond within the review period in ER-17.01 with one of: code 1 (no comments), code 2 (proceed

subject to comments), code 3 (revise and resubmit) or code 4 (for information). Work shall not start on items subject to review until code 1 or 2 has been obtained.

6.3 Errors in the Contractor's documents

Errors, omissions or ambiguities in the Contractor's documents shall be corrected at the Contractor's cost, notwithstanding any review.

6.4 As-built documents and database

The Contractor shall deliver as-built documents, the 3D model and the final engineering database in accordance with ER-17.03 as a condition of the Performance Certificate.

6.5 Intellectual property

Copyright and other intellectual property in the Contractor's documents remain with the Contractor or its suppliers. The Owner receives a non-exclusive, perpetual, royalty-free licence to use them for the Project. Proprietary source code of turbine control systems remains with the supplier and is placed in escrow.

7 Staff and Labour

7.1 Engagement of staff

The Contractor shall engage staff and labour in compliance with Turkish labour law, social security law and work permit requirements, and shall pay wages and benefits not lower than those established for the trade in the region.

7.2 Key personnel

The Contractor shall provide the key personnel named in its tender (project director, engineering manager, construction manager, commissioning manager, HSE manager, quality manager) and shall not replace them without the Owner's consent.

7.3 Accommodation and welfare

The Contractor shall provide accommodation, transport and welfare facilities for its personnel in accordance with applicable standards and the IFC/EBRD worker accommodation guidance.

8 Plant, Materials and Workmanship

8.1 Manner of execution

Plant and materials shall be new, of the quality specified and suitable for the intended purpose. Workmanship shall be carried out in a proper and workmanlike manner.

8.2 Inspection

The Engineer may inspect, examine and test materials and workmanship during manufacture and construction at any reasonable time. The Contractor shall give notice when work is ready and before it is covered up.

8.3 Testing

The Contractor shall perform the tests specified in the Contract, including FATs listed in ER-15.03. The Engineer shall be given at least 15 days' notice of witness points.

8.4 Rejection and remedial work

If plant, materials or workmanship are found defective, the Engineer may reject them and the Contractor shall promptly remedy the defect and ensure that the rejected item complies with the Contract.

8.5 Ownership of plant and materials

Title in plant and materials passes to the Owner on the earlier of their delivery to Site or payment for them, free of liens and encumbrances. Risk remains with the Contractor under Clause 17 until Taking-Over.

9 Commencement, Delays and Programme

9.1 Commencement

The Contractor shall commence the Works on the NTP date and proceed with due expedition and without delay.

9.2 Time for Completion and Key Dates

The Contractor shall complete the Works within the Time for Completion (975 days from NTP) and achieve the Key Dates in Appendix G, in particular first fire, first synchronisation in combined cycle and Taking-Over.

9.3 Programme

The Contractor shall submit a Level 3 CPM programme within 28 days after NTP in accordance with ER-17.02 and update it monthly. The accepted programme does not relieve the Contractor of any obligation.

9.4 Extension of Time

The Contractor is entitled to an extension of the Time for Completion if and to the extent that completion is or will be delayed by: a Variation; an Exceptional Event under Clause 19; unforeseeable physical conditions under Sub-Clause 4.5; a delay caused by the Owner or its other contractors; or a change in law under Sub-Clause 14.4; in each case subject to Clause 22. Concurrent delays caused by the Contractor are taken into account.

9.5 Delay damages

If the Contractor fails to achieve Taking-Over within the Time for Completion, it shall pay delay damages of 0.1 % of the Contract Price per day of delay. Delay damages are capped at 10 % of the Contract Price. If first synchronisation in combined cycle is delayed by more than 60 days beyond its Key Date, interim delay damages of 0.05 % per day apply, to be credited against final delay damages.

9.6 Rate of progress

If progress is too slow to complete within the Time for Completion, the Engineer may instruct the Contractor to submit a revised programme and to accelerate at the Contractor's cost.

10 Tests on Completion

10.1 Contractor's obligations

The Contractor shall carry out the Tests on Completion described in ER-16 after providing the as-built documents and O&M manuals required for testing. Test procedures shall be submitted 90 days before the tests.

10.2 Delayed tests

If the Owner unduly delays the Tests on Completion for more than 30 days, the Contractor may give notice and the Works are deemed to be taken over, subject to the tests being carried out when possible.

10.3 Retesting

If the Works fail any Test on Completion, the Contractor shall remedy the failure and repeat the test at its cost, including the cost of fuel above the quantities allowed for the first test.

10.4 Failure to pass Tests on Completion

If the Works fail to pass the performance tests after the rectification period in Sub-Clause 12.5, the consequences in Clause 12 apply.

11 Taking-Over

11.1 Conditions for Taking-Over

The Engineer shall issue the Taking-Over Certificate when all of the following are fulfilled: the Works are complete except for minor punch items not affecting safe operation; all Tests on Completion have been passed or performance liquidated damages have been paid; the Reliability Run has been passed; emission and environmental guarantees are met; the required permits for operation are available; and training has been completed.

11.2 No partial taking-over

The Works shall be taken over as a whole. Operation of the gas turbine in open cycle during commissioning does not constitute taking-over, and the energy produced belongs to the Owner.

11.3 Punch list

Punch items listed at Taking-Over shall be completed within 90 days. The Owner may withhold twice the estimated value of open punch items from the Taking-Over milestone payment.

12 Performance Guarantees and Liquidated Damages

12.1 Guarantees

The Contractor guarantees that the Works achieve the values in Appendix B when tested in accordance with the Employer's Requirements.

12.2 Performance liquidated damages

If the Works fail to achieve a guaranteed value for which a liquidated damages rate is stated in Appendix B, the Contractor shall pay liquidated damages at that rate, pro rata for fractions.

12.3 Minimum acceptance levels

If the net output is below 97 % of the guaranteed value or the net heat rate is above 103 % of the guaranteed value after the rectification period, the Owner may reject the Works, in which case Sub-Clause 20.2 applies.

12.4 Must-meet guarantees

The emission, noise near-field and environmental guarantees marked "must rectify" in Appendix B are absolute. The Contractor shall rectify the Works until they are met; Taking-Over shall not be issued before compliance.

12.5 Rectification period

The Contractor may, within 180 days after the first performance test, carry out modifications and repeat the test at its cost. The Owner may operate the plant commercially during this period.

12.6 Caps

Performance liquidated damages are capped at 15 % of the Contract Price. The aggregate of delay and performance liquidated damages is capped at 20 % of the Contract Price.

12.7 Nature of liquidated damages

The Parties agree that the liquidated damages are a genuine pre-estimate of the Owner's loss and are the Owner's sole financial remedy for the failure concerned, without prejudice to the Owner's rights under Sub-Clauses 12.3, 12.4 and 20.2.

13 Defects Liability

13.1 Defects Notification Period

The Defects Notification Period is 24 months from Taking-Over. For marine works, civil structures and buildings it is 36 months.

13.2 Remedying defects

The Contractor shall remedy any defect notified during the Defects Notification Period at its cost within the time reasonably stated by the Owner, including dismantling, transport and re-installation. If the Contractor fails to do so, the Owner may carry out the work at the Contractor's cost.

13.3 Extension of the Defects Notification Period

The Defects Notification Period of a part of the Works is extended by the period during which that part cannot be used because of a defect, and restarts for 12 months for replaced parts, up to a maximum of 48 months from Taking-Over.

13.4 Serial defects

If the same defect occurs in three or more identical components, it is deemed a serial defect and the Contractor shall replace or modify all identical components, including those not yet failed.

13.5 Performance Certificate

The Engineer shall issue the Performance Certificate within 28 days after the latest expiry of the Defects Notification Periods, when all defects have been remedied and the as-built documentation has been delivered.

14 Variations and Adjustments

14.1 Right to vary

The Owner may instruct Variations at any time before Taking-Over. The Contractor shall execute them unless it promptly demonstrates that it cannot readily obtain the required goods or that the Variation would affect safety or the guarantees.

14.2 Value engineering

The Contractor may propose changes that reduce cost, accelerate completion or improve efficiency. If accepted, the net saving is shared equally between the Parties.

14.3 Variation procedure

Before a Variation is instructed, the Engineer may request a proposal. The Contractor shall respond within 28 days with a description, programme impact and price based on the rates in the Contract or, where none apply, on reasonable cost plus 10 % overhead and profit.

14.4 Change in law

The Contract Price and Time for Completion shall be adjusted for changes in Turkish law after the Base Date that affect the Contractor's cost or time, excluding changes to corporate income tax of the Contractor.

15 Contract Price and Payment

15.1 Contract Price

The Contract Price is the lump sum stated in Article A4. It includes all duties, taxes and fees payable by the Contractor except Value Added Tax, which the Owner pays in addition where applicable, and customs duties on imported permanent plant, which are exempted under the Owner's investment incentive certificate.

15.2 Advance payment

The Owner shall pay an advance of 10 % of the Contract Price within 30 days after receiving the Advance Payment Guarantee and an invoice. The advance is recovered pro rata from milestone payments MS-02 to MS-10.

15.3 Milestone payments

The Contractor may invoice each milestone in Appendix C when it has been achieved and certified by the Engineer. No payment is due for partially achieved milestones.

15.4 Retention

The Owner shall retain 5 % of each milestone payment. Half of the retention is released at Taking-Over and half at the Performance Certificate; alternatively the retention may be released against a Retention Guarantee of equal value.

15.5 Time for payment

The Owner shall pay each certified amount within 45 days after receiving the Contractor's invoice with supporting documents.

15.6 Delayed payment

Late payments bear interest at EURIBOR (3 months) plus 3 % per annum, calculated daily.

15.7 Currency

Payments are made in euros to the accounts designated by the Leader for the offshore and onshore portions respectively.

16 Securities

16.1 Securities to be provided

The Contractor shall provide: the Performance Security (Sub-Clause 4.2); an Advance Payment Guarantee equal to the advance, reducing as it is recovered; a Retention Guarantee if retention is released early; and a parent company guarantee from the ultimate parent of each consortium member covering its obligations under the Contract.

16.2 Claims under securities

The Owner may claim under a security only for amounts to which it is entitled under the Contract and shall give the Contractor 7 days' notice before a claim, except where the security is about to expire.

17 Care of the Works and Indemnities

17.1 Care of the Works

The Contractor bears full responsibility for the care of the Works from NTP until Taking-Over, and thereafter for any outstanding work.

17.2 Owner's risks

The Owner bears the risk of loss or damage caused by: war, hostilities, rebellion or civil war; ionising radiation not caused by the Contractor; pressure waves from aircraft; use or occupation of part of the Works by the Owner; and design provided by the Owner.

17.3 Indemnities

The Contractor shall indemnify the Owner against claims for death, injury, loss or damage to property arising from the execution of the Works, except to the extent caused by the negligence of the Owner. The Owner shall indemnify the Contractor against claims arising from the Owner's risks.

18 Insurance

18.1 Construction and erection all risks

The Contractor shall procure a construction and erection all risks (CEAR) policy covering the Works for their full replacement value, including testing and commissioning, with the Owner and its lenders as co-insured and the lenders as loss payee.

18.2 Third party liability

The Contractor shall maintain third party liability insurance of not less than EUR 50,000,000 per occurrence.

18.3 Marine cargo

The Contractor shall insure all plant shipped to Site under a marine cargo policy on a warehouse-to-site basis.

18.4 Delay in start-up

The Owner may procure delay in start-up (DSU) insurance. The Contractor shall cooperate with the insurers and comply with their reasonable risk-management recommendations.

19 Exceptional Events

19.1 Definition

An Exceptional Event is an event beyond a Party's control which it could not reasonably have provided against, could not reasonably have avoided or overcome, and which is not substantially attributable to the other Party, including war, terrorism, epidemics declared by public authorities, natural catastrophes such as earthquakes of intensity exceeding the design basis, and nationwide strikes. Earthquakes within the design basis of the Works are not Exceptional Events.

19.2 Notice

The affected Party shall give notice within 14 days after becoming aware of the Exceptional Event and shall use all reasonable endeavours to minimise the delay.

19.3 Consequences

The Contractor is entitled to an extension of time for delays caused by an Exceptional Event and, for war, terrorism and riots in Türkiye, to its reasonable costs.

19.4 Prolonged Exceptional Event

If the execution of substantially all the Works is prevented for a continuous period of 180 days, either Party may terminate the Contract by 30 days' notice. The Owner shall then pay for work done and the cost of demobilisation.

20 Suspension and Termination

20.1 Suspension by the Owner

The Owner may suspend the Works at any time. If the suspension exceeds 84 days not caused by the Contractor, the Contractor is entitled to an extension of time and costs, and may treat the suspension as an omission of the affected part after a further 28 days' notice.

20.2 Termination for Contractor's default

The Owner may terminate the Contract by 14 days' notice if the Contractor: fails to provide the Performance Security; abandons the Works; fails without reasonable excuse to proceed; reaches the cap of delay damages; fails the minimum acceptance levels in Sub-Clause 12.3; becomes insolvent (both members); or commits bribery. The Owner may then complete the Works through others and recover the additional cost, and may reject the Works and recover all amounts paid where the plant is incapable of commercial use.

20.3 Termination for convenience

The Owner may terminate the Contract for convenience by 56 days' notice. The Owner shall then pay for work done, goods ordered, demobilisation costs and 5 % of the value of the work not executed as compensation for loss of profit.

20.4 Termination by the Contractor

The Contractor may terminate the Contract if the Owner fails to pay a certified amount within 90 days after its due date, or suspends the Works for more than 180 days without cause, after giving 14 days' notice to remedy.

21 Limitation of Liability

21.1 Total liability

The total liability of the Contractor under or in connection with the Contract shall not exceed 100 % of the Contract Price.

21.2 Indirect loss

Neither Party is liable to the other for loss of use, loss of profit, loss of contract or any indirect or consequential loss, except as expressly provided by the liquidated damages in Clauses 9 and 12.

21.3 Exceptions

The limitations in this Clause do not apply in case of fraud, gross negligence, wilful misconduct, abandonment of the Works, or to the indemnities for death and personal injury and third-party property damage.

22 Claims and Disputes

22.1 Claims

A Party that considers itself entitled to additional time or payment shall give notice within 28 days after becoming aware of the event, and a fully detailed claim within 84 days. Failure to give timely notice results in the loss of the entitlement, except where the other Party was not prejudiced.

22.2 Dispute Avoidance/Adjudication Board

A standing three-member Dispute Avoidance/Adjudication Board shall be appointed within 42 days after NTP. It visits the Site every four months, assists in avoiding disputes and gives decisions on referred disputes within 84 days. Decisions are binding unless and until revised by arbitration.

22.3 Amicable settlement

Before arbitration the Parties shall attempt to settle the dispute through meetings of senior executives within 28 days after a notice of dissatisfaction with a Board decision.

22.4 Arbitration

Disputes not settled shall be finally resolved under the Rules of Arbitration of the International Chamber of Commerce by three arbitrators, seated in Istanbul, in English. The Works shall continue during any dispute.

23 Miscellaneous

23.1 Anti-bribery and compliance

Each Party shall comply with applicable anti-bribery laws, export control and sanctions regulations. Breach by the Contractor entitles the Owner to terminate under Sub-Clause 20.2.

23.2 Assignment

Neither Party may assign the Contract without the other's consent, except that the Owner may assign its rights by way of security to its lenders. The Contractor shall sign a direct agreement with the lenders in customary form.

23.3 Entire agreement and amendments

The Contract is the entire agreement between the Parties about its subject matter. Amendments are valid only if in writing and signed by both Parties; each amendment shall be recorded in the Owner's Database.

23.4 Severability

If a provision is held invalid, the remaining provisions remain in force and the Parties shall replace the invalid provision with a valid one of the closest economic effect.

Appendix A - Employer's Requirements (technical)

The requirements below are mandatory ("shall"). Each states its verification method and, where applicable, the acceptance criterion and the related guarantee in Appendix B.

ER-01 General requirements and plant description

ER-01.01 Purpose of the Works

The Contractor shall design, engineer, procure, manufacture, deliver, construct, erect, commission, test and hand over a complete, safe and fully operational natural-gas-fired combined cycle power plant of 600 MW net output at Site Reference Conditions (the Works), ready for commercial operation in the Turkish electricity market.

Discipline: general | Verification: review | Acceptance: Taking-Over Certificate issued

ER-01.02 Plant configuration

The plant shall be a 1x1 multi-shaft combined cycle block comprising one heavy-duty H-class (or equivalent advanced class) gas turbine with its own generator, one triple-pressure reheat heat recovery steam generator (HRSG) with selective catalytic reduction (SCR) and CO catalyst, and one reheat condensing steam turbine with its own generator, cooled by once-through seawater.

Discipline: process | Verification: review | Acceptance: Configuration as described in the approved Basic Design Package

ER-01.03 Design life and operating regime

All equipment and structures shall be designed for an economic life of 30 years, 8,000 equivalent operating hours per year and up to 250 starts per year (of which up to 50 cold, 100 warm and 100 hot starts), including daily cycling and load following.

Discipline: general | Verification: analysis | Acceptance: Design life documented in design reports and OEM statements

ER-01.04 Turnkey completeness

The Works shall include everything necessary for safe, reliable and efficient operation whether or not expressly mentioned in the Contract (fit-for-purpose obligation), except the items expressly allocated to the Owner in Appendix D (scope split).

Discipline: general | Verification: review

ER-01.05 Units and language

SI units shall be used throughout (pressure in bar(g)/bar(a), temperature in degC). All documents, drawings and labels shall be in English; operator-facing labels and safety signs shall be bilingual English/Turkish.

Discipline: general | Verification: review

ER-01.06 Plant identification system

All systems, equipment, instruments, cables and documents shall be identified according to KKS (VGB-B 105/106). The tag data shall be classified according to the AVEVA class library adopted by the Owner and delivered through the Owner's engineering database.

Discipline: general | Verification: review | Acceptance: 100 % of delivered tags classified and validated in the Owner's database

ER-01.07 Codes and standards hierarchy

The Works shall comply with (in order of precedence): Turkish laws and regulations; the Grid Code of the TSO; the Contract; the codes and standards listed in Appendix A.19; and good industry practice. Where codes conflict the more stringent requirement applies.

Discipline: general | Verification: review

ER-01.08 Proven technology

Major equipment (GT, ST, generators, HRSG, GSU transformers) shall be of a design with at least two units of the same frame in commercial operation for not less than 8,000 fired hours each at the date of the Contract, or supported by a validation programme accepted by the Owner.

Discipline: general | Verification: review | Acceptance: Reference list accepted by the Owner

ER-01.09 Design responsibility

The Contractor is fully responsible for the design, including the verification of all Owner-provided data. Review or approval of documents by the Owner or the Engineer does not relieve the Contractor of any obligation.

Discipline: general | Verification: review

ER-01.10 Engineering database and data handover

The Contractor shall deliver all engineering data (equipment, lines, instruments, cables, documents, design parameters) in the Owner's engineering database format, kept current at each document issue, and hand over native CAD/3D models (Plant 3D or E3D) and IFC4 exports.

Discipline: general | Verification: review | Acceptance: Data handover accepted at Taking-Over

ER-02 Site conditions and design data

ER-02.01 Site location and area

The Site is a 12.0 ha plot (400 m east-west x 300 m north-south) on the northern Marmara coast, levelled by the Owner to +15.0 m, plant grid origin at the south-west corner, plant north = true north. The usable area inside the 10 m stream-channel buffer on the west boundary and the 5 m landscaping strip is about 10.6 ha (E 20-395, N 12-295); the 60 m public coastal strip lies between the plot and the shoreline. A further 5.04 ha temporary construction area north of the plot (E 0-180, N 320-600) is provided by the Owner (tie-in TP-A1). Details and existing features per ALP-OWN-GEN-001 and drawings ALP-OWN-DWG-001 (external connections) and ALP-OWN-DWG-002 (site and construction area plan).

Discipline: civil | Verification: inspection

ER-02.02 Ambient design range

Equipment shall operate at full capability between -8 degC and +40 degC dry bulb and 18 % to 100 % relative humidity, and survive -15 degC to +45 degC. Climatic data per ALP-OWN-MET-001 (10-year site record; extremes -11.4 degC / +39.6 degC).

Discipline: process | Verification: analysis

ER-02.03 Site Reference Conditions (SRC)

Guarantees refer to: ambient dry-bulb temperature 15.0 degC; relative humidity 70 %; barometric pressure 1011.5 mbar; seawater temperature at intake 16.0 degC (contractual reference; the measured annual mean of 15.4 degC in ALP-OWN-MAR-001 applies to the design range only); natural gas per ER-03.02 reference composition; grid frequency 50.0 Hz; power factor 0.85 lagging.

Discipline: process | Verification: review

ER-02.04 Seawater conditions

Seawater quality, temperature and marine conditions per ALP-OWN-MAR-001 (two-layer sea, pycnocline at 20 m to 25 m). Design range at the intake depth 6.9 degC to 27.0 degC, chloride up to 14,500 mg/l, TSS up

to 140 mg/l in storms, occasional H₂S in the lower layer, mussel settlement, jellyfish blooms up to 20 kg/m³ and mucilage events.

Discipline: process | Verification: analysis

ER-02.05 Wind, snow and rain

Basic wind speed 30 m/s (10 min mean, 10 m height, 50-year return); ground snow load 0.75 kN/m²; design rainfall intensity 110 mm/h (10 min, 25-year return) for drainage design.

Discipline: civil | Verification: analysis

ER-02.06 Seismic design

The Site is in a high seismic hazard region. Structures and equipment shall be designed per the Turkish Building Earthquake Code (TBDY 2018) with a site-specific hazard study; design earthquake DD-2 (475-year) peak ground acceleration 0.45 g; building importance factor 1.5 for power block structures; functional recovery of critical systems after DD-3 events.

Discipline: structural | Verification: analysis | Acceptance: Seismic design report approved; site-specific hazard study delivered

ER-02.07 Geotechnical conditions

Owner-provided baseline investigation indicates alluvial deposits over weathered rock at 18 m to 25 m depth, groundwater at +13.5 m and liquefaction potential in the upper sand layers. The Contractor shall perform confirmatory investigations and select foundation solutions (piling or ground improvement) accordingly.

Discipline: civil | Verification: analysis

ER-02.08 Corrosive coastal atmosphere

Outdoor equipment and structures shall be designed for corrosivity category C5 (ISO 12944-2) with a coating system of high durability (H, more than 15 years to first major maintenance).

Discipline: mechanical | Verification: inspection

ER-02.09 Flood protection

Finished floor levels of all buildings and equipment foundations shall be at least 0.5 m above the 100-year coastal flood level including a sea-level-rise allowance of 0.5 m.

Discipline: civil | Verification: analysis

ER-02.10 Airborne salt and dust

The plant shall be designed for the airborne contaminants in ALP-OWN-MET-001 section 6: sea-salt aerosol up to 165 ug/m³ (as NaCl) during lodos storms, chloride deposition up to 420 mg/m² per day, PM₁₀ up to 260 ug/m³. The gas turbine inlet air filtration shall keep sodium plus potassium at the compressor inlet within the OEM limit for all these conditions, including high-humidity and fog (coalescer/droplet separator stage upstream of the fine filters, hydrophobic final filters of class E12 or better). Outdoor electrical equipment: creepage distance for very heavy pollution (IEC 60815, 31 mm/kV).

Discipline: mechanical | Verification: analysis | Acceptance: Filter design report with OEM confirmation (Na+K at compressor inlet below OEM limit at 165 ug/m³ NaCl); Na+K air sampler downstream of the filters with 12 months of data reported during the Defects Notification Period

ER-02.11 Design ambient points

HVAC and cooling design: summer 0.4 % 32.8 degC / 48 % RH; freeze protection and heating: winter 99.6 % -4.8 degC; performance test ambient range 5 degC to 30 degC (ALP-OWN-MET-001 sections 3 and 7). Heat tracing for all outdoor water-filled lines and instruments.

Discipline: hvac | Verification: review

ER-02.12 Lightning

Lightning protection per IEC 62305 with a risk assessment based on 26 thunderstorm days per year and ground flash density 2.4 flashes/km²/year (ALP-OWN-MET-001 section 5).

Discipline: electrical | Verification: analysis

ER-02.13 Existing features on the Site

The Contractor shall demolish and remove the abandoned warehouse foundations (E 300-360, N 20-60), protect the Owner's meteorological mast and groundwater wells, and keep clear of the buried 11 kV cable along the east boundary until the Owner has had it relocated (ALP-OWN-GEN-001 section 2).

Discipline: civil | Verification: inspection

ER-02.14 Access and transport

The Contractor is responsible for the transport of all plant to Site, including route surveys, permits for abnormal loads and temporary works, taking into account the 120 t bridge limits on the state road (ALP-OWN-GEN-001 section 3). Heavy components shall be landed through a temporary barge landing on the public coastal strip, designed, built, operated and removed by the Contractor; the Owner obtains the coastal permit (tie-in TP-M1).

Discipline: civil | Verification: review

ER-03 Fuel

ER-03.01 Fuels

Primary fuel: pipeline natural gas per ALP-OWN-FUL-001. Secondary (back-up) fuel: light diesel oil (LDO) per ALP-OWN-FUL-002, for up to 500 hours per year. The gas turbine shall be dual-fuel with automatic transfer between fuels.

Discipline: process | Verification: review

ER-03.02 Natural gas specification

The plant shall operate on the full natural gas range of ALP-OWN-FUL-001 sections 2 to 4: CH₄ 85.0 to 98.5 mol %, C₂H₆ up to 7.0 %, C₃H₈ up to 2.5 %, C₄ up to 1.0 %, C₅+ up to 0.2 %, N₂ up to 4.0 %, CO₂ up to 2.5 %, O₂ up to 0.02 %; LHV 44.2 MJ/kg to 50.0 MJ/kg (emission limits apply over the whole range; performance guarantees at the reference gas); Wobbe index (GCV) 47.2 MJ/Sm³ to 51.8 MJ/Sm³ with rate of change up to 1 %/min. Hydrogen admixture up to 5 vol % without modification and with all guarantees; provisions for 10 vol % by future OEM retrofit (fuel system materials, gas detection, control system capacity). Reference composition for guarantees: CH₄ 93.0, C₂H₆ 3.5, C₃H₈ 0.8, i-C₄ 0.15, n-C₄ 0.15, C₅+ 0.05, N₂ 1.55, CO₂ 0.8 mol %; LHV 48.4 MJ/kg.

Discipline: process | Verification: analysis | Acceptance: OEM fuel specification compliance statement for the full range

ER-03.03 Gas supply conditions at terminal point

Pressure at the terminal point TP-G1: normal 55-65 barg, minimum 45 barg for full load, exceptional 40 barg for up to 72 h/year; temperature 5-25 degC (ALP-OWN-FUL-001 section 6). No gas compressor is provided: between 40 and 45 barg the gas turbine operates on gas at the pressure-limited load (about 60 % GT load at 40 barg); below 40 barg the control system transfers automatically to LDO (ER-03.10).

Discipline: process | Verification: analysis

ER-03.04 Gas receiving and fiscal metering station

2x100 % filtration/separation, 2x100 % water-bath or electric heaters, 2x100 % pressure-reduction streams with active/monitor regulators and slam-shut valves; fiscal metering with 2x100 % ultrasonic meters (OIML R137 accuracy class 0.5), on-line gas chromatograph and flow computers complying with the network operator's requirements.

Discipline: mechanical | Verification: inspection

ER-03.05 Performance fuel gas heating

The GT fuel gas shall be preheated with intermediate-pressure feedwater to the OEM-optimised temperature (about 200 degC) to maximise combined cycle efficiency; a start-up heater shall be provided.

Discipline: process | Verification: review

ER-03.06 Natural gas contaminant limits

At the gas turbine fuel skid inlet the gas shall meet the OEM limits and at least: total sulphur ≤ 30 mg S/Sm³; H₂S + COS ≤ 5 mg/Sm³; mercury ≤ 1 ug/Sm³; sodium + potassium, lead, vanadium and calcium within the OEM trace-metal limits (network data below 0.005 mg/Sm³, not guaranteed by the network operator); no free liquids; particulates removed to the OEM limit (coalescing filter-separators, 3 um absolute final filtration); superheat at least 28 K above the hydrocarbon and water dew points at all loads. The Contractor shall provide continuous monitoring of H₂S and total sulphur and periodic sampling points for trace metals and mercury.

Discipline: process | Verification: test | Acceptance: Gas conditioning design and commissioning samples within limits

ER-03.07 Light diesel oil specification and burner limits

LDO as delivered per ALP-OWN-FUL-002 (TS EN 590 based: density 820-845 kg/m³, viscosity 2.0-4.5 mm²/s at 40 degC, flash point ≥ 55 degC, sulphur ≤ 10 mg/kg, water ≤ 200 mg/kg, ash ≤ 0.01 %, FAME ≤ 7 %, LHV 42.6-43.2 MJ/kg). At the gas turbine fuel inlet: Na + K ≤ 0.5 mg/kg, vanadium ≤ 0.5 mg/kg, lead ≤ 1.0 mg/kg, calcium ≤ 2.0 mg/kg, water ≤ 100 mg/kg, filtration to 5 um, or stricter OEM limits.

Discipline: process | Verification: test | Acceptance: Fuel samples at GT inlet within limits during LDO tests

ER-03.08 LDO unloading and storage

Road tanker unloading station with 4 bays (metering, strainers, spill containment, earthing), and two vertical fixed-roof storage tanks each of 3,500 m³ net working volume (together 72 hours of base-load operation on LDO; heel and fire-fighting reserves in addition), in a bund of 110 % of the largest tank, with foam fire protection per NFPA 11 and NFPA 30, level/leak monitoring and water draw-off. Operation beyond the storage autonomy is not required (resupply is limited to 25 trucks/day).

Discipline: mechanical | Verification: inspection

ER-03.09 LDO forwarding and treatment

2 x 100 % forwarding pumps, duplex filters, flow metering, and a mandatory fuel treatment system of 2 x 100 % centrifugal separators with water washing, sized to reduce Na+K from 1.8 mg/kg to below the ER-03.07 limit at the gas turbine inlet.

Discipline: mechanical | Verification: test

ER-03.10 Fuel transfer and LDO start

Automatic transfer from gas to LDO on loss of gas supply pressure, from any load between minimum environmental load and base load, without unit trip; manual on-load transfer back to gas; black start is not

required but start-up on LDO with grid power shall be possible. Purging and liquid fuel system recirculation to avoid coking during gas operation.

Discipline: i_and_c | Verification: demonstration | Acceptance: Automatic transfer demonstrated at 50 % and 100 % load | Guarantee: PG-18

ER-03.11 NOx control and SCR on LDO

NOx on LDO shall be controlled by water injection (demineralised water) and the SCR; the SCR and ammonia system shall be designed for LDO operation (sulphur 10 mg/kg, ammonium bisulphate risk at the cold end).

Discipline: process | Verification: test | Guarantee: PG-17

ER-04 Performance and operational requirements

ER-04.01 Net electrical output

Guaranteed net electrical output at SRC: not less than 600.0 MW at the 380 kV terminal point.

Discipline: process | Verification: test | Acceptance: ≥ 600.0 MW corrected to SRC | Guarantee: PG-01

ER-04.02 Net heat rate

Guaranteed net heat rate at SRC: not more than 5,860 kJ/kWh (LHV), corresponding to a net efficiency of about 61.4 %.

Discipline: process | Verification: test | Acceptance: $\leq 5,860$ kJ/kWh corrected to SRC | Guarantee: PG-02

ER-04.03 Part-load efficiency

Net heat rate at 60 % plant load shall not exceed 6,250 kJ/kWh (LHV) at SRC.

Discipline: process | Verification: test | Acceptance: $\leq 6,250$ kJ/kWh | Guarantee: PG-03

ER-04.04 Start-up times

From GT ignition to base load: hot start (shutdown < 8 h) ≤ 45 min; warm start (8 h to 48 h) ≤ 90 min; cold start (> 48 h) ≤ 180 min, without exceeding the life consumption limits agreed with the OEM.

Discipline: process | Verification: demonstration | Acceptance: Hot start ≤ 45 min demonstrated three times | Guarantee: PG-09

ER-04.05 Load ramp rate

Combined cycle load ramp rate of at least 35 MW/min between minimum environmental load and base load.

Discipline: i_and_c | Verification: demonstration | Acceptance: ≥ 35 MW/min | Guarantee: PG-10

ER-04.06 Minimum environmental load

Stable emissions-compliant operation shall be possible down to 40 % of plant net output or lower.

Discipline: process | Verification: demonstration | Acceptance: ≤ 40 % | Guarantee: PG-11

ER-04.07 Frequency and voltage support

The plant shall provide primary and secondary frequency control, fast frequency response and reactive power support as required by the Grid Code, including operation within 47.5 Hz to 52.5 Hz for the durations specified therein.

Discipline: electrical | Verification: test | Acceptance: Grid Code compliance tests accepted by the TSO

ER-04.08 Reactive capability

Each generator shall be capable of power factor 0.85 lagging to 0.95 leading at rated active power at the generator terminals.

Discipline: electrical | Verification: test

ER-04.09 House load and island operation

The unit shall survive a full load rejection to house load and remain in stable operation for at least 60 minutes.

Discipline: i_and_c | Verification: demonstration

ER-04.10 Auxiliary power

Auxiliary power consumption at base load shall not exceed 2.2 % of gross output; the value is included in the net output guarantee.

Discipline: electrical | Verification: test

ER-04.11 Degradation

The Contractor shall submit recoverable and non-recoverable degradation curves for output and heat rate over 48,000 EOH; they apply for test corrections.

Discipline: process | Verification: review

ER-04.12 Availability target

The design shall support an equivalent availability factor of at least 93 % averaged over the first six years, considering the OEM maintenance programme.

Discipline: general | Verification: analysis

ER-05 Gas turbine

ER-05.01 Gas turbine type

Heavy-duty, single-shaft, 50 Hz, H-class (or equivalent) gas turbine with dry low-NOx combustion capable of the fuel range in ER-03.02.

Discipline: mechanical | Verification: review

ER-05.02 Inlet air system

Three-stage static air filtration (coastal environment, final stage EPA E12 per EN 1822), anti-icing, evaporative cooler provision, silencer.

Discipline: mechanical | Verification: inspection

ER-05.03 Combustion dynamics monitoring

On-line combustion dynamics monitoring with automatic tuning shall be provided.

Discipline: i_and_c | Verification: demonstration

ER-05.04 Enclosure and ventilation

Acoustic enclosure with ventilation, gas detection and fire suppression suitable for the hazardous area classification.

Discipline: mechanical | Verification: inspection

ER-05.05 Compressor washing

On-line and off-line compressor washing systems with detergent skid.

Discipline: mechanical | Verification: inspection

ER-05.06 Condition monitoring and remote diagnostics

Vibration, performance and combustion monitoring with secure remote diagnostic access for the OEM (IEC 62443 compliant).

Discipline: i_and_c | Verification: review

ER-05.07 Maintenance intervals

Hot gas path inspection interval shall be not less than 25,000 EOH and major inspection not less than 50,000 EOH at the operating regime of ER-01.03.

Discipline: mechanical | Verification: review

ER-06 Heat recovery steam generator

ER-06.01 HRSG type

Horizontal-gas-flow, triple-pressure, reheat, natural-circulation HRSG, designed for fast cycling (Benson or equivalent HP section acceptable).

Discipline: mechanical | Verification: review

ER-06.02 Design code

Pressure parts per ASME Section I or EN 12952, PED 2014/68/EU conformity assessment by a notified body.

Discipline: mechanical | Verification: inspection | Acceptance: PED certificates delivered

ER-06.03 SCR and CO catalyst

In-duct SCR with aqueous ammonia (< 25 %) injection and an oxidation catalyst sized for the emission guarantees at end of catalyst life (24,000 h).

Discipline: process | Verification: test | Guarantee: PG-04

ER-06.04 Ammonia system

Aqueous ammonia storage (2x7 days at base load), unloading, vapourisation and dosing with leak detection and bunding.

Discipline: process | Verification: inspection

ER-06.05 Cycling capability

The HRSG shall withstand the start regime of ER-01.03 without drum or header fatigue beyond design life (fatigue analysis per EN 12952-3).

Discipline: mechanical | Verification: analysis

ER-06.06 Stack

Steel stack 65 m high (EIA decision ALP-OWN-ENV-001), with CEMS platform, sampling ports per EN 15259 and aviation lighting.

Discipline: structural | Verification: inspection

ER-06.07 Drains and blowdown

Continuous and intermittent blowdown, HRSG drains recovery to the condenser where water quality permits.

Discipline: process | Verification: review

ER-06.08 Preservation

Dry and wet lay-up systems (nitrogen capping and dehumidified air) for long shutdowns.

Discipline: process | Verification: inspection

ER-07 Steam turbine and condenser

ER-07.01 Steam turbine

Reheat condensing steam turbine with HP, IP and LP sections, axial or side exhaust, 50 Hz, with HP/IP/LP bypass systems of 100 % capacity.

Discipline: mechanical | Verification: review

ER-07.02 Main steam conditions

HP live steam about 170 bar(a)/600 degC and hot reheat about 600 degC at base load, as optimised by the OEM.

Discipline: process | Verification: review

ER-07.03 Condenser

Single-shell or two-shell surface condenser with titanium tubes (Grade 2, welded) and titanium-clad tube sheets, divided water boxes for on-load cleaning of one half, and an automatic ball-type tube cleaning system.

Discipline: mechanical | Verification: inspection

ER-07.04 Cooling water temperature rise

Temperature rise across the condenser shall not exceed 7 K at base load.

Discipline: process | Verification: test | Acceptance: ≤ 7 K | Guarantee: PG-13

ER-07.05 Vacuum system

2x100 % liquid-ring vacuum pumps (holding) plus hogging capability for fast start-up.

Discipline: mechanical | Verification: inspection

ER-07.06 Lube and control oil

Fire-resistant control fluid, lube oil system with 2x100 % AC pumps, DC emergency pump and oil purification.

Discipline: mechanical | Verification: inspection

ER-07.07 Condenser design standard

Thermal design per HEI Standards for Steam Surface Condensers (11th edition or later): cleanliness factor 0.85, 5 % tube plugging margin, tube velocity 1.8 m/s to 2.2 m/s, titanium wall per HEI. Hotwell storage at least 3 minutes of maximum condensate flow; dissolved oxygen at hotwell outlet ≤ 20 ug/l in normal operation.

Discipline: process | Verification: analysis | Acceptance: Condenser datasheet and HEI calculation approved

ER-07.08 Condenser pressure

Design condenser pressure at base load and Site Reference Conditions (seawater 16.0 degC, 7 K rise): not more than 36 mbar(a). At the maximum seawater temperature (27.0 degC) the plant shall run at base load without load limitation; the back-pressure shall stay within the steam turbine continuous limits with margin to the alarm value.

Discipline: process | Verification: test | Acceptance: Design requirement (condenser datasheet and HEI calculation); contractual compliance through PG-02; ASME PTC 12.2 test for information

ER-07.09 Steam bypass to the condenser

The condenser shall accept 100 % HP/IP/LP bypass steam at the maximum steam flow (including cold-ambient and LDO cases) with desuperheating, dump-device design per HEI, and protection against shell overheating and tube erosion.

Discipline: mechanical | Verification: analysis

ER-07.10 Condenser leak detection and protection

Cation conductivity measurement per water box half and in the hotwell with automatic alarm and isolation logic for seawater ingress; water boxes rubber-lined with impressed-current cathodic protection; air in-leakage allowance and vacuum pump sizing per HEI.

Discipline: i_and_c | Verification: test

ER-07.11 Circulating water flow

Circulating water flow for the design heat duty at 7 K rise (indicatively 43,000 m³/h to 47,000 m³/h, to be confirmed by the Contractor); CW system hydraulic transient analysis including pump trip and valve closure.

Discipline: process | Verification: analysis

ER-08 Cooling water system**ER-08.01 Once-through seawater cooling**

Offshore intake head above the pycnocline at about -12 m (at least 3 m above seabed, about 650 m offshore) with velocity cap and intake velocity ≤ 0.15 m/s; intake pipelines; pump house; outfall diffuser at least 250 m from the intake discharging into the upper layer (ALP-OWN-MAR-001 section 6).

Discipline: civil | Verification: inspection

ER-08.02 Screening

Bar screens, 2x100 % travelling band screens with fish return system, trash rakes.

Discipline: mechanical | Verification: inspection

ER-08.03 Circulating water pumps

2x50 % vertical CW pumps with a common spare pump assembly stored on Site; materials suitable for seawater (duplex).

Discipline: mechanical | Verification: test

ER-08.04 Biofouling control

Electro-chlorination with intermittent/shock dosing; residual chlorine at outfall ≤ 0.1 mg/l.

Discipline: process | Verification: test | Acceptance: ≤ 0.1 mg/l residual chlorine

ER-08.05 Thermal plume

The outfall shall limit the temperature excess to ≤ 3 K at the edge of the 100 m mixing zone.

Discipline: process | Verification: test | Acceptance: ≤ 3 K | Guarantee: PG-14

ER-08.06 Closed cooling water

Closed cooling water circuit for auxiliaries with 2x100 % plate heat exchangers (titanium plates) and 2x100 % pumps.

Discipline: mechanical | Verification: inspection

ER-08.07 Jellyfish and mucilage

Screens, trash rakes and screen wash shall remain operable at jellyfish loads up to 20 kg/m³ and during mucilage events; the CW system shall allow operation at reduced flow with automatic load reduction rather than trip.

Discipline: mechanical | Verification: demonstration

ER-09 Water treatment and effluents

ER-09.01 Raw water

Seawater reverse osmosis (2 x 100 %) with ultrafiltration pre-treatment (raw SDI₁₅ > 5), designed for the upper-layer quality of ALP-OWN-MAR-001 including boron removal for demineralisation feed.

Discipline: process | Verification: test

ER-09.02 Demineralised water

Two-pass RO plus electrodeionisation, 2 x 100 %, conductivity < 0.1 uS/cm, silica < 10 ug/l, with capacity for continuous water injection at 50 % load on LDO; demineralised water storage 2 x 1,250 m³ (3 days of cycle make-up plus about 24 h of water injection); connection point for mobile demin trailers.

Discipline: process | Verification: test

ER-09.03 Cycle chemistry

Water/steam chemistry per VGB-S-010 (all-volatile treatment, oxygenated where applicable) with on-line analysers and sampling panel.

Discipline: process | Verification: review

ER-09.04 Condensate polishing

Full-flow condensate polishing for start-up and seawater ingress events.

Discipline: process | Verification: inspection

ER-09.05 Waste water

Neutralisation, oil separation and treatment of process wastewater, and a packaged biological treatment plant with UV disinfection for sanitary wastewater (no municipal sewer), so that all effluents meet the Turkish Water Pollution Control Regulation sea-discharge limits before the outfall.

Discipline: hse | Verification: test | Acceptance: Effluent within permit limits

ER-09.06 Stormwater

Separate clean and potentially oily stormwater drainage with oil interceptors.

Discipline: civil | Verification: inspection

ER-10 Balance of plant mechanical systems

ER-10.01 Feed water pumps

2x100 % variable-speed (VFD or hydraulic coupling) HP/IP boiler feed pumps.

Discipline: mechanical | Verification: test

ER-10.02 Condensate pumps

2x100 % condensate extraction pumps.

Discipline: mechanical | Verification: test

ER-10.03 Compressed air

2x100 % oil-free compressors, 2x100 % dryers, instrument air dew point -40 degC, receiver for 15 min of instrument air after loss of compressors.

Discipline: mechanical | Verification: test

ER-10.04 Auxiliary boiler

Gas-fired auxiliary boiler for start-up and sealing steam, sized for a cold start without external steam.

Discipline: mechanical | Verification: test

ER-10.05 Fire protection

Fire detection and protection per NFPA 850 and Turkish Fire Regulation: ring main, 2x100 % electric plus diesel fire pumps, water mist/CO2 for GT enclosure, deluge for transformers, gas detection, fire alarm system.

Discipline: hse | Verification: test | Acceptance: Insurer and fire brigade acceptance

ER-10.06 HVAC

HVAC for all buildings; electrical and control rooms with N+1 cooling and positive pressure with filtered (coastal) air.

Discipline: hvac | Verification: test

ER-10.07 Cranes

Turbine hall EOT crane sized for the heaviest maintenance lift (excluding generator stator), and maintenance hoists for pumps and valves.

Discipline: mechanical | Verification: test

ER-10.08 Nitrogen and hydrogen

Nitrogen supply for purging and preservation; hydrogen supply for generator cooling if hydrogen-cooled generators are offered.

Discipline: mechanical | Verification: inspection

ER-11 Electrical systems

ER-11.01 Grid connection

Connection to the 380 kV transmission system via an indoor GIS switchyard in a ventilated building (salt atmosphere), double busbar and bus coupler, two line bays and two generator transformer bays, designed for 50 kA short-circuit current; outdoor bushings with creepage per ER-02.10.

Discipline: electrical | Verification: test

ER-11.02 Generators

Air- or hydrogen-cooled synchronous generators per IEC 60034, 50 Hz, static excitation with PSS, generator circuit breakers per IEC/IEEE 62271-37-013.

Discipline: electrical | Verification: test

ER-11.03 Generator step-up transformers

One GSU per generator, ONAN/ONAF, rated for the maximum generator output at -8 degC, on-load tap changer, per IEC 60076; fire walls between transformers.

Discipline: electrical | Verification: test

ER-11.04 Auxiliary power supply

Two unit auxiliary transformers, 10.5 kV MV switchboards (2 sections with bus tie), 0.4 kV PMCC with automatic transfer. Start-up power from the grid via back-feed through the GSU.

Discipline: electrical | Verification: test

ER-11.05 Emergency power

Emergency diesel generator for safe shutdown loads (turning gear, lube oil, emergency lighting); the diesel fire pump has its own independent drive; UPS (2x100 %) for DCS and critical loads with 60 min autonomy; 220 V DC and 24 V DC systems.

Discipline: electrical | Verification: test

ER-11.06 Protection

Duplicated main protection (Main 1/Main 2 from different manufacturers where required by the TSO) for generators, transformers, busbars and lines.

Discipline: electrical | Verification: test

ER-11.07 Earthing and lightning

Earthing grid per IEEE 80 with safe touch and step voltages; lightning protection per IEC 62305.

Discipline: electrical | Verification: test

ER-11.08 Cables

Cables per IEC standards, halogen-free flame-retardant inside buildings, fire-rated cables for safety circuits.

Discipline: electrical | Verification: inspection

ER-11.09 Metering and TSO interface

Revenue metering per TSO/market operator rules (main and check meters, class 0.2S), RTU and telecommunication to the national dispatch centre.

Discipline: electrical | Verification: test

ER-12 Instrumentation, control and cybersecurity

ER-12.01 Distributed control system

Single integrated plant DCS for GT, ST, HRSG and BOP with redundant controllers, networks and power supplies; operation of the whole plant from the central control room by two operators.

Discipline: i_and_c | Verification: test

ER-12.02 Automation level

Automatic start-up and shutdown sequences (one-button start to base load) and automatic load control from the dispatch set-point.

Discipline: i_and_c | Verification: demonstration

ER-12.03 Safety instrumented systems

SIS per IEC 61511 with SIL verification, including burner management and HRSG protection.

Discipline: i_and_c | Verification: analysis | Acceptance: SIL verification reports approved

ER-12.04 CEMS

Certified continuous emission monitoring (NOx, CO, O2, NH3, flow, moisture) per EN 14181 with data transmission to the environmental authority.

Discipline: i_and_c | Verification: test

ER-12.05 Historian and performance monitoring

Plant historian (5 years on-line storage) and on-line performance calculation per ASME PTC 46 principles.

Discipline: i_and_c | Verification: test

ER-12.06 Operator training simulator

High-fidelity operator training simulator of the whole plant, delivered 6 months before first fire.

Discipline: i_and_c | Verification: demonstration

ER-12.07 Cybersecurity

OT cybersecurity per IEC 62443 (target security level SL2), zones and conduits, DMZ, patch and backup management, security acceptance test.

Discipline: i_and_c | Verification: test | Acceptance: Security acceptance test passed

ER-12.08 Time synchronisation

GPS time synchronisation of DCS, protection relays and recorders (1 ms sequence of events).

Discipline: i_and_c | Verification: test

ER-13 Civil, structural and architectural works

ER-13.01 Foundations

Turbine and generator foundations designed for dynamic loads with separated foundations from buildings; settlement limits per OEM.

Discipline: civil | Verification: analysis

ER-13.02 Buildings

Turbine hall, control and electrical building, administration building, workshop/warehouse, water treatment building, gatehouse, fire station.

Discipline: civil | Verification: inspection

ER-13.03 Structural steel

Structural steel per EN 1993 with seismic detailing, hot-dip galvanised outdoors (EN ISO 1461).

Discipline: structural | Verification: inspection

ER-13.04 Concrete

Concrete per EN 1992 / TS 500 with exposure class XS for marine-exposed elements; minimum 50-year durability.

Discipline: civil | Verification: inspection

ER-13.05 Roads and drainage

Internal asphalt roads for 40 t axle loads to the heaviest delivery route, surface water drainage for ER-02.05 rainfall.

Discipline: civil | Verification: inspection

ER-13.06 Marine works

Intake and outfall pipelines (HDPE or GRP) with ballast and scour protection, designed for 50-year wave conditions.

Discipline: civil | Verification: analysis

ER-14 Environment, health and safety

ER-14.01 EIA compliance

The Works shall comply with the conditions of the positive EIA decision and the Environmental Law (No. 2872) and its regulations.

Discipline: hse | Verification: review | Acceptance: No outstanding EIA non-conformities at Taking-Over

ER-14.02 Air emissions

Natural gas: NO_x ≤ 30 mg/Nm³, CO ≤ 30 mg/Nm³, NH₃ slip ≤ 5 mg/Nm³. LDO: NO_x ≤ 50 mg/Nm³, CO ≤ 50 mg/Nm³, SO₂ ≤ 10 mg/Nm³, dust ≤ 5 mg/Nm³, NH₃ slip ≤ 5 mg/Nm³. All dry, 15 % O₂, 50 % to 100 % GT load (ALP-OWN-ENV-001 section 1).

Discipline: hse | Verification: test | Acceptance: Within guaranteed limits | Guarantee: PG-04

ER-14.03 Near-field noise

Maximum 85 dB(A) at 1 m from any equipment in normal operation.

Discipline: hse | Verification: test | Acceptance: ≤ 85 dB(A) | Guarantee: PG-07

ER-14.04 Environmental noise

Maximum 45 dB(A) LAeq night at the nearest sensitive receptor.

Discipline: hse | Verification: test | Acceptance: ≤ 45 dB(A) | Guarantee: PG-08

ER-14.05 Occupational health and safety

Construction and design shall comply with the Occupational Health and Safety Law (No. 6331); the Contractor shall implement an ISO 45001-certified HSE management system and a lost-time-injury target of zero.

Discipline: hse | Verification: review

ER-14.06 Hazardous areas

Hazardous area classification per IEC 60079-10-1; equipment certified accordingly.

Discipline: hse | Verification: review

ER-14.07 Process safety studies

HAZID, HAZOP, SIL determination, fire and explosion risk assessment and a Major Accident Prevention assessment where applicable.

Discipline: hse | Verification: review | Acceptance: All actions closed before first fire

ER-14.08 Carbon capture readiness

Space reservation and tie-in points for a future post-combustion carbon capture plant at the HRSG outlet.

Discipline: process | Verification: review

ER-15 Quality, inspection and testing during manufacture

ER-15.01 Quality management

ISO 9001 quality management for the Contractor and all major subcontractors; project quality plan within 30 days of NTP.

Discipline: general | Verification: review

ER-15.02 Inspection and test plans

ITPs for all major equipment with hold and witness points for the Engineer, 15 days' notice for witnessed tests.

Discipline: general | Verification: review

ER-15.03 Factory acceptance tests

FATs at least for: GT and ST control systems, generators, GSU transformers, GIS, MV switchgear, DCS, CEMS.

Discipline: general | Verification: test

ER-15.04 Material traceability

Material certificates per EN 10204 3.1 (3.2 for pressure parts above 50 bar) and positive material identification of alloy piping.

Discipline: mechanical | Verification: inspection

ER-15.05 Welding

Welding per ASME IX / EN ISO 15614, 100 % NDE of HP/HRH/main steam butt welds.

Discipline: piping | Verification: inspection

ER-16 Commissioning and Tests on Completion

ER-16.01 Commissioning plan

System-based commissioning plan with pre-commissioning, cold and hot commissioning, submitted 9 months before first fire.

Discipline: commissioning | Verification: review

ER-16.02 Steam blowing and chemical cleaning

Chemical cleaning of the HRSG and steam blowing of steam lines to OEM cleanliness criteria.

Discipline: commissioning | Verification: test

ER-16.03 Grid Code compliance tests

All tests required by the TSO for connection and commercial operation, including reactive capability and frequency response.

Discipline: electrical | Verification: test | Acceptance: TSO acceptance

ER-16.04 Performance test

Performance test per ASME PTC 46 within 30 days after initial operation, test procedure submitted 90 days before, with instrument calibration traceable to national standards.

Discipline: commissioning | Verification: test | Acceptance: PG-01/PG-02/PG-03 demonstrated | Guarantee: PG-01

ER-16.05 Emission and noise tests

Emission tests by an accredited laboratory and noise surveys per PG-04 to PG-08.

Discipline: hse | Verification: test | Guarantee: PG-04

ER-16.06 Operational demonstration

Demonstration of start times, ramp rates, minimum load, load rejection and trip tests.

Discipline: commissioning | Verification: demonstration | Guarantee: PG-09

ER-16.07 Reliability run

30-day continuous reliability run with availability $\geq 98\%$; restarted after rectification if failed.

Discipline: commissioning | Verification: test | Acceptance: $\geq 98\%$ | Guarantee: PG-12

ER-16.08 Tests on LDO

Performance test on LDO (24 h, corrected to SRC), emission tests on LDO, automatic fuel transfer tests and a start on LDO.

Discipline: commissioning | Verification: test | Acceptance: PG-15, PG-16, PG-17, PG-18 demonstrated | Guarantee: PG-15

ER-17 Documentation, training, O&M and spares

ER-17.01 Document management

Documents submitted electronically with the Owner's numbering and a master document register; review period 14 days (21 days for the Basic Design Package).

Discipline: project_control | Verification: review

ER-17.02 Schedule and reporting

Level 3 CPM schedule (MS Project XML or Primavera P6) with monthly updates, S-curves and progress reports.

Discipline: project_control | Verification: review

ER-17.03 As-built documentation

As-built drawings, 3D model and engineering database within 3 months of Taking-Over.

Discipline: general | Verification: review

ER-17.04 O&M manuals

Operation and maintenance manuals (English, operating procedures also in Turkish) 6 months before first fire, final at Taking-Over.

Discipline: general | Verification: review

ER-17.05 Training

Classroom, simulator and on-the-job training for 60 Owner staff (operators, maintenance, I&C, chemistry), minimum 12 weeks per key trade.

Discipline: general | Verification: demonstration

ER-17.06 Spare parts

Commissioning spares, special tools, and a recommended two-year operating spares list with prices valid for 18 months.

Discipline: general | Verification: review

ER-17.07 Long-term service

The power island supplier shall offer a long-term service agreement for GT/ST/generators (separate contract, not part of the Contract Price).

Discipline: general | Verification: review

ER-18 Interfaces and terminal points

ER-18.01 Terminal points

Terminal points and scope limits are defined in Appendix D; the Contractor shall provide all works up to and including the terminal points.

Discipline: general | Verification: review

ER-18.02 Grid connection interface

The Owner builds and energises the 380 kV line up to TP-E1 (available by 2028-07-15, an Owner-furnished item under Sub-Clause 2.4) and coordinates the line energisation with the TSO. The Contractor coordinates plant-side protection settings, SCADA and the Grid Code compliance tests with the TSO and the Owner's line contractor.

Discipline: electrical | Verification: review

ER-18.03 Gas network interface

The Contractor shall coordinate the gas receiving station tie-in, pressure protection and metering approval with the gas network operator.

Discipline: process | Verification: review

ER-18.04 Owner's engineering database

All interface data shall be exchanged through the Owner's engineering database (single source of truth).

Discipline: general | Verification: review

ER-18.05 Tie-in register

The tie-in points, their location, conditions and the Owner's availability dates are listed in the tie-in register (Appendix D). The Contractor's scope starts at each tie-in; the Contractor shall verify the conditions and coordinate the interface works.

Discipline: general | Verification: review

Appendix B - Performance Guarantees and Liquidated Damages

ID	Guarantee	Value	Conditions	Test	Liquidated damages / remedy	Minimum acceptance
PG-01	Net electrical output at Site Reference Conditions	>= 600 MW	Base load, new and clean (corrected for recoverable degradation per agreed curves), Site Reference Conditions (ER-02.03), power factor 0.85 lagging at the 380 kV terminal point, all auxiliaries in normal operation.	ASME PTC 46 overall plant performance test, corrections to Site Reference Conditions per agreed correction curves; measured at the 380 kV revenue meters.	2,200 EUR per kW shortfall. Below 97 % of the guaranteed value the Owner may reject the Works after the rectification period (Sub-Clause 12.5).	582 MW
PG-02	Net heat rate (LHV) at Site Reference Conditions	<= 5860 kJ/kWh	As PG-01, fuel within the Fuel Specification (ER-03), heat input on LHV basis.	ASME PTC 46, fuel flow by fiscal ultrasonic meters, LHV by on-line gas chromatograph (ISO 6976).	400,000 EUR per kJ/kWh excess. Above 103 % of the guaranteed value the Owner may reject the Works after the rectification period.	6035.8 kJ/kWh
PG-03	Net heat rate (LHV) at 60 % plant load	<= 6250 kJ/kWh	Plant at 60 % of net output, Site Reference Conditions, sliding pressure.	ASME PTC 46, part-load test point.	150,000 EUR per kJ/kWh excess	-
PG-04	NOx emission (as NO2)	<= 30 mg/Nm3	Dry flue gas, 15 % O2, 50 % to 100 % GT load, all ambient conditions within the design range.	EN 14792 via certified CEMS (EN 14181 QAL2) plus independent stack test.	No LD: the Contractor shall rectify at its cost until compliant; Taking-Over is not issued before compliance.	-
PG-05	CO emission	<= 30 mg/Nm3	Dry flue gas, 15 % O2, 50 % to 100 % GT load.	EN 15058 via CEMS plus stack test.	Must rectify (as PG-04).	-
PG-06	Ammonia slip from the SCR	<= 5 mg/Nm3	Dry flue gas, 15 % O2, 50 % to 100 % GT load, catalyst new.	Stack test per recognised method (e.g. EN ISO 21877).	Must rectify.	-
PG-07	Near-field noise	<= 85 dB(A)	1 m from equipment surfaces / enclosures, 1.5 m height, all operating modes except emergency.	ISO 3744 / ISO 3746 survey.	Must rectify.	-
PG-08	Noise at nearest sensitive receptor (night)	<= 45 dB(A)	LAeq at the nearest residential receptor defined in the EIA, plant at base load, night period.	Environmental noise measurement per ISO 1996-2 with background correction.	250,000 EUR per dB(A) exceedance (after rectification attempts). Contractor to rectify first; LD only if rectification fails.	-
PG-09	Hot start time to base load	<= 45 min	Shutdown shorter than 8 h, GT ignition to plant base load, emissions compliant from minimum environmental load.	Demonstration during Tests on Completion, three successful starts.	20,000 EUR per minute exceedance	-
PG-10	Load ramp rate in combined cycle	>= 35 MW/min	Between minimum environmental load and base load, emissions compliant.	Demonstration, recorded by DCS historian.	100,000 EUR per MW/min shortfall	-
PG-11	Minimum environmental load	<= 40 % of plant net output	Lowest load with PG-04/PG-05 compliance at Site Reference Conditions.	Demonstration with CEMS records.	300,000 EUR per percentage point exceedance	-
PG-12	Reliability run availability	>= 98 %	30 consecutive days of operation following Owner dispatch, maximum two forced outages, none longer than 48 h.	Reliability Run per ER-16.07.	Failed run to be repeated from the start after rectification.	-
PG-13	Condenser cooling water temperature rise	<= 7 K	Base load, all seawater temperatures within the design range.	Temperature measurement at condenser inlet/outlet.	Must rectify (EIA permit condition).	-
PG-14	Thermal plume at edge of mixing zone	<= 3 K	Temperature excess over ambient seawater at 100 m from the outfall diffuser, base load, summer conditions.	Thermal plume survey per EIA monitoring plan.	Must rectify (EIA permit condition).	-
PG-15	Net electrical output on LDO at Site Reference Conditions	>= 560 MW	Base load on the reference LDO (LHV 42.9 MJ/kg, FAME 0 %, water 100 mg/kg), water injection in service, SRC; corrections from	ASME PTC 46, 24 h LDO test; fuel flow by Coriolis meters, LHV per ASTM D240.	800 EUR per kW shortfall	543.2 MW

ID	Guarantee	Value	Conditions	Test	Liquidated damages / remedy	Minimum acceptance
			the actual test fuel by agreed curves (TQ-023).			
PG-16	Net heat rate (LHV) on LDO at Site Reference Conditions	<= 6150 kJ/kWh	Base load on the reference LDO (LHV 42.9 MJ/kg, FAME 0 %, water 100 mg/kg), water injection in service, SRC; corrections from the actual test fuel by agreed curves (TQ-023).	ASME PTC 46.	60,000 EUR per kJ/kWh excess	6334.5 kJ/kWh
PG-17	NOx emission on LDO (as NO2)	<= 50 mg/Nm3	Dry, 15 % O2, 50 % to 100 % GT load on LDO.	EN 14792 via CEMS and stack test.	No LD: the Contractor shall rectify until compliant (permit condition).	-
PG-18	Automatic fuel transfer gas to LDO without trip	>= 100 % successful transfers	Three transfers at 50 % load and three at base load, triggered by simulated loss of gas pressure.	Demonstration during Tests on Completion.	Must rectify; Taking-Over not issued before successful demonstration.	-

Caps: performance liquidated damages 15 % and aggregate liquidated damages 20 % of the Contract Price (Sub-Clause 12.6). Delay damages per Sub-Clause 9.5.

Appendix C - Schedule of Payments

ID	Milestone	Payment trigger	Planned month after NTP	Share	Amount	Security
MS-01	Advance payment	Contract effective and receipt of the Advance Payment Guarantee	0	10 %	EUR 53,248,000	Advance Payment Guarantee 10 %
MS-02	Approval of basic design and power island order	Owner approval of the Basic Design Package and evidence of firm power island sub-orders	3	8 %	EUR 42,598,400	
MS-03	Gas turbine and generator manufacturing progress	Evidence of GT rotor forging and generator stator core completion (inspection report)	9	12 %	EUR 63,897,600	
MS-04	Main civil works started	Completion of GT/ST foundation piling and first mat pour	8	5 %	EUR 26,624,000	
MS-05	Delivery of gas turbine to Site	GT package delivered to Site, receiving inspection passed	16	12 %	EUR 63,897,600	
MS-06	Delivery of steam turbine, generators and GSU transformers	Main components delivered to Site, receiving inspection passed	18	10 %	EUR 53,248,000	
MS-07	HRSg pressure parts erected and hydrotested	Successful HRSg hydrostatic test witnessed by the Engineer	20	8 %	EUR 42,598,400	
MS-08	Back-feed energisation	380 kV switchyard and GSU energised from the grid, auxiliary power available	21	5 %	EUR 26,624,000	
MS-09	First fire of the gas turbine	GT first fire on natural gas	23	7 %	EUR 37,273,600	
MS-10	First synchronisation in combined cycle	Steam turbine synchronised to the grid	26	8 %	EUR 42,598,400	
MS-11	Taking-Over of the Works	Issue of the Taking-Over Certificate (after Tests on Completion and Reliability Run)	32	10 %	EUR 53,248,000	Retention release against Retention Guarantee
MS-12	Final completion	Issue of the Performance Certificate at the end of the Defects Notification Period	56	5 %	EUR 26,624,000	
	Total			100 %	EUR 532,480,000	

Amounts are computed from the Contract Price; the advance is recovered pro rata from MS-02 to MS-10 and 5 % retention applies to each milestone (Clause 15).

Appendix D - Scope of Works Split and Terminal Points

Party codes: IEC = Imaginary Electric Company; IEPC = Istanbul EPC Mühendislik ve Taahhüt A.Ş.; OWNER = Alpilla Enerji Üretim A.Ş.

ID	Area	Item	Design	Supply	Install	Commission	Terminal point
SC-001	Site	Land, access rights and site handover (levelled to +15.0 m)	OWNER	OWNER	OWNER	-	Site boundary per land-use plan
SC-002	Site	Environmental Impact Assessment (EIA) positive decision	OWNER	OWNER	-	-	
SC-003	Site	Construction, operating and building permits (Contractor supplies all design inputs)	IEPC	OWNER	-	-	
SC-004	Site	Geotechnical and topographic surveys (baseline supplied by Owner, confirmatory by Contractor)	IEPC	IEPC	IEPC	-	
SC-005	Site	Site preparation, fencing, roads, drainage, temporary facilities	IEPC	IEPC	IEPC	-	
SC-006	Power island	Gas turbine package incl. auxiliaries, enclosure, inlet air filtration, exhaust diffuser	IEC	IEC	IEPC	IEC	GT exhaust flange to HRSG inlet duct
SC-007	Power island	GT generator incl. excitation, protection, cooling	IEC	IEC	IEPC	IEC	Generator terminals (IPB connection)
SC-008	Power island	Steam turbine (HP/IP/LP, reheat) incl. lube/control oil, gland steam, turning gear	IEC	IEC	IEPC	IEC	
SC-009	Power island	ST generator incl. excitation, protection	IEC	IEC	IEPC	IEC	Generator terminals
SC-010	Power island	HRSG triple-pressure reheat, horizontal, natural circulation, with SCR and CO catalyst	IEC	IEC	IEPC	IEC	HRSG outlet to stack
SC-011	Power island	Stack 60 m with CEMS platform and silencer	IEPC	IEPC	IEPC	IEPC	
SC-012	Power island	Water-cooled condenser (seawater), vacuum system	IEC	IEC	IEPC	IEC	Cooling water flanges at condenser water boxes
SC-013	Power island	Power island control system (TCS/ST controls) integrated in plant DCS	IEC	IEC	IEPC	IEC	DCS interface (redundant fibre)
SC-014	Power island	GSU transformers (GT and ST), isolated phase busducts, generator circuit breakers	IEC	IEC	IEPC	IEC	GSU HV bushings
SC-015	Balance of plant	Condensate and boiler feed water systems (pumps 2x100 %)	IEPC	IEPC	IEPC	IEPC	
SC-016	Balance of plant	Seawater intake, screens, CW pumps (2x50 %), outfall and diffuser	IEPC	IEPC	IEPC	IEPC	Offshore intake head / outfall diffuser
SC-017	Balance of plant	Electro-chlorination for biofouling control	IEPC	IEPC	IEPC	IEPC	
SC-018	Balance of plant	Seawater reverse osmosis and demineralisation plant	IEPC	IEPC	IEPC	IEPC	
SC-019	Balance of plant	Condensate polishing plant	IEPC	IEPC	IEPC	IEPC	
SC-020	Balance of plant	Waste water treatment and effluent monitoring	IEPC	IEPC	IEPC	IEPC	Discharge point to outfall
SC-021	Balance of plant	Closed cooling water system with plate heat exchangers	IEPC	IEPC	IEPC	IEPC	
SC-022	Balance of plant	Compressed air (service and instrument) 2x100 %	IEPC	IEPC	IEPC	IEPC	
SC-023	Balance of plant	Auxiliary boiler (gas-fired) for start-up steam	IEPC	IEPC	IEPC	IEPC	
SC-024	Balance of plant	Fire detection and protection per NFPA 850	IEPC	IEPC	IEPC	IEPC	
SC-025	Balance of plant	HVAC for all buildings and electrical rooms	IEPC	IEPC	IEPC	IEPC	
SC-026	Balance of plant	Cranes and hoists (turbine hall EOT crane)	IEPC	IEPC	IEPC	IEPC	
SC-027	Fuel gas	Gas pipeline from national transmission system to plant boundary	OWNER	OWNER	OWNER	OWNER	Owner valve at plant boundary (inlet flange of gas receiving station)

ID	Area	Item	Design	Supply	Install	Commission	Terminal point
SC-028	Fuel gas	Gas receiving and metering station (fiscal metering, filtration, heating, pressure control)	IEPC	IEPC	IEPC	IEPC	Inlet flange at plant boundary
SC-029	Fuel gas	Performance gas heater and GT fuel skid	IEC	IEC	IEPC	IEC	
SC-030	Electrical	380 kV GIS switchyard (double busbar) within plant boundary	IEPC	IEPC	IEPC	IEPC	380 kV line gantry / cable sealing ends
SC-031	Electrical	380 kV transmission connection line and remote substation works	OWNER	OWNER	OWNER	OWNER	Line gantry at switchyard
SC-032	Electrical	Unit auxiliary transformers, MV switchgear 10.5 kV, LV 0.4 kV distribution	IEPC	IEPC	IEPC	IEPC	
SC-033	Electrical	Emergency diesel generator and UPS / DC systems	IEPC	IEPC	IEPC	IEPC	
SC-034	Electrical	Earthing, lightning protection, cabling, cable trays	IEPC	IEPC	IEPC	IEPC	
SC-035	Electrical	Revenue metering and TSO SCADA/telecom interface	IEPC	IEPC	IEPC	IEPC	TSO RTU interface in control building
SC-036	I&C	Plant DCS, historian, operator training simulator	IEPC	IEPC	IEPC	IEPC	
SC-037	I&C	Safety instrumented systems (burner management, HRSG protection) per IEC 61511	IEC	IEC	IEPC	IEC	
SC-038	I&C	Continuous emission monitoring system (CEMS)	IEPC	IEPC	IEPC	IEPC	Data link to environmental authority
SC-039	I&C	Plant security, CCTV, access control, cybersecurity (IEC 62443)	IEPC	IEPC	IEPC	IEPC	
SC-040	Civil	Piling, foundations, structures, turbine hall, HRSG structure	IEPC	IEPC	IEPC	-	
SC-041	Civil	Buildings: control, administration, workshop/warehouse, water treatment, gatehouse	IEPC	IEPC	IEPC	-	
SC-042	Civil	Marine works for intake and outfall (dredging, pipes, heads)	IEPC	IEPC	IEPC	-	
SC-043	Utilities	Potable water connection at site boundary (no sewer available)	OWNER	OWNER	OWNER	-	Site boundary
SC-044	Utilities	Telecommunication connection (fibre) to site boundary	OWNER	OWNER	OWNER	-	Site boundary
SC-045	Utilities	Construction power and water	IEPC	IEPC	IEPC	-	
SC-046	Commissioning	Consumables until Taking-Over (except natural gas)	IEPC	IEPC	-	-	
SC-047	Commissioning	Natural gas for commissioning and tests	OWNER	OWNER	-	-	Gas receiving station inlet
SC-048	Commissioning	Grid access for back-feed and test energy; purchase of test energy	OWNER	OWNER	-	-	
SC-049	Commissioning	Operation and maintenance staff for training and commissioning participation	OWNER	OWNER	-	-	
SC-050	Spares	Commissioning spares and special tools	IEPC	IEPC	-	-	
SC-051	Spares	Two-year operating spares (priced list, Owner option)	IEPC	IEPC	-	-	Owner's warehouse
SC-052	Fuel oil	LDO truck unloading station (4 bays), storage tanks 2 x 50 %, bund, foam system	IEPC	IEPC	IEPC	IEPC	Tanker coupling at unloading bays
SC-053	Fuel oil	LDO forwarding pumps, filters, metering, fuel treatment (2 x 100 % centrifugal separators)	IEPC	IEPC	IEPC	IEPC	GT liquid fuel skid inlet flange
SC-054	Power island	Dual-fuel GT combustion system, liquid fuel skid, water injection skid, purge system	IEC	IEC	IEPC	IEC	
SC-055	Commissioning	LDO first fill of the storage tanks	OWNER	OWNER	-	-	Tanker coupling at unloading bays
SC-056	Commissioning	LDO for tests beyond the first fill	-	IEPC	-	-	
SC-057	Balance of plant	Packaged sanitary wastewater treatment plant (biological + UV), discharge via outfall	IEPC	IEPC	IEPC	IEPC	Discharge into the outfall

Tie-in register

ID	Service	Location	Size	Conditions	Owner provides	Contractor provides	Available by
TP-A1	Temporary construction area (5 ha)	E 0-180, N 320-600 (5.04 ha), north of the plot, west of the access road			Lease of the 5 ha field until Taking-Over, free of charge to the Contractor	Laydown, pre-assembly, contractor facilities; restoration at the end	2027-02-01
TP-A2	Site (plot) handover	Plot E 0-400, N 0-300 (plant grid), levelled to +15.0 m		East strip E 370-400 available 2027-06-30 (needed by the Contractor 2027-09-01 for the gas station civil works).	Plot free of encumbrances, levelled to +15.0 m, boundary monuments; relocation of the buried 11 kV cable (E 390) by 2027-06-30	All works within the plot from the access date	2026-11-02
TP-CW1	Seawater intake head	About 650 m offshore at -12 m	about 2 x DN2200 (Contractor to size)	6.9-27.0 degC	Seabed survey and marine permit	Intake head, pipelines, pump house	2027-04-30
TP-CW2	Outfall diffuser	About 350 m offshore, at least 250 m from the intake			Marine permit	Outfall pipeline and diffuser	2027-04-30
TP-D1	Storm water discharge	Western stream channel (max 2 m ³ /s) or to sea			Water authority consent	Site drainage, oil interceptors, outlet structure	2026-11-02
TP-E1	Grid connection 380 kV	Line gantry at the north-west corner	380 kV, 2 circuits	380 kV nominal, 420 kV highest; 360-410 kV normal, 342-420 kV up to 30 min; 49.0-51.0 Hz; short-circuit 12-40 kA (2029), 50 kA (2035 planning); connection capacity 640 MW export / 40 MW import (ALP-OWN-ELE-001).	Double-circuit 380 kV OHL up to the dead-end tower, conductors to the gantry	Gantry, GIS switchyard, protection and metering	2028-07-15
TP-E2	Construction power	North gate	34.5 kV / 5 MVA	34.5 kV +/- 10 %, 5 MVA, prospective short-circuit 12.5 kA (distribution company).	34.5 kV connection, 5 MVA, metering	Construction substations and distribution	2027-02-28
TP-G1	Natural gas supply	North-east plot corner, flange at EL +14.0	16 in (DN400), CL600	Normal 55-65 barg; minimum 45 barg for full load; exceptional 40 barg up to 72 h/year (GT load limited, ER-03.03); MOP 70 / MAOP 75 barg; 5-25 degC; 115,000 Sm ³ /h firm (density 0.7325 kg/Sm ³ at 15 degC, reference gas).	16-inch spur line (4.5 km), tap valve station, isolation valve and insulating joint	Gas receiving and metering station from the inlet flange	2028-06-30
TP-L1	LDO supply	North gate (road tankers)	4 in tanker coupling (API RP 1004 bottom loading / dry-break)	Up to 25 trucks/day (about 30 m ³ each); 4 bays at about 100 m ³ /h each unloading pump set; LDO per TS EN 590 based specification.	LDO supply contract (up to 25 trucks/day), first fill	Unloading station and storage	2028-09-01
TP-M1	Temporary barge landing	Public coastal strip south of the plot		Heaviest pieces: GT generator stator 340 t, GT GSU 268 t (transport weights, IEC data).	Coastal permit: application by 2027-04-30 if the Contractor's permit design is received by 2027-03-31; permit targeted by 2027-09-30	Design, construction, operation and removal of the landing; marine works only 1 October - 30 April	2027-09-30

ID	Service	Location	Size	Conditions	Owner provides	Contractor provides	Available by
TP-R1	Road access	North gate		6 m asphalt, state road bridges limited to 120 t gross vehicle weight; components above about 90 t via the barge landing TP-M1 (ALP-OWN-GEN-001 section 3).	Existing 6 m asphalt road to the state road	Site roads and gate	2026-11-02
TP-S1	Sanitary wastewater	None - no municipal sewer available			Nothing (no sewer)	Packaged sanitary treatment plant (SC-057); treated effluent to the outfall	-
TP-T1	Telecommunication	North gate			Fibre ducts of two operators to the plot boundary	Site telecom network and SCADA links	2027-06-30
TP-T2	Emission data link to the environmental authority	CEMS data logger in the control building			Registration of the plant with the Ministry's continuous emission monitoring centre, credentials, connection agreement	CEMS, data acquisition and the data link (SC-038) up to the Ministry server interface	2028-08-01
TP-T3	TSO SCADA and revenue metering link	TSO RTU and metering cubicle in the control building			TSO signal list and protocol, TSO end of the link, OPGW on the Owner's line	RTU, revenue meters, fibre from the gantry joint box, telecom equipment (SC-035)	2028-05-01
TP-W1	Potable water	North gate	DN150	2.5-6.0 barg at the meter outlet, 20 m3/h (municipal supply), potable quality per the Turkish drinking water regulation.	DN150 municipal main connection and meter, 3 barg, 20 m3/h	Potable water storage and distribution	2027-02-28

Appendix E - Site and Design Data

Category	Parameter	Value	Unit	Condition	Basis
ambient	Ambient temperature - minimum for full capability	-8	degC		ER-02.02
ambient	Ambient temperature - maximum for full capability	40	degC		ER-02.02
ambient	Ambient temperature - survival range	-15 to +45	degC		ER-02.02
ambient	SRC ambient dry-bulb temperature	15	degC	Site Reference Conditions	ER-02.03
ambient	SRC relative humidity	70	%	Site Reference Conditions	ER-02.03
ambient	SRC barometric pressure	1011.5	mbar	Site Reference Conditions	ER-02.03
ambient	Summer design dry bulb (0.4 %)	32.8	degC	coincident RH 48 %	SRC-OWNR-0001, ER-02.11
ambient	Winter design dry bulb (99.6 %)	-4.8	degC	coincident RH 88 %	SRC-OWNR-0001, ER-02.11
ambient	Record maximum / minimum temperature	39.6 / -11.4	degC	1960-2024	SRC-OWNR-0001
ambient	Annual mean temperature (measured)	14.9	degC	2015-2024 site mast	SRC-OWNR-0001
ambient	Sea-salt aerosol maximum (event)	165	ug/m3	as NaCl, lodos storms	SRC-OWNR-0001, ER-02.10
ambient	Sea-salt aerosol annual mean	14	ug/m3	as NaCl	SRC-OWNR-0001
ambient	Chloride deposition maximum	420	mg/m2/day	ISO 9225, category S3	SRC-OWNR-0001, ER-02.10
ambient	PM10 maximum (event)	260	ug/m3	Saharan dust episodes	SRC-OWNR-0001,

Category	Parameter	Value	Unit	Condition	Basis
					ER-02.10
ambient	Ground flash density	2.4	flashes/km2/year	26 thunderstorm days/year	SRC-OWNR-0001, ER-02.12
ambient	Performance test ambient range	5 to 30	degC		SRC-OWNR-0001, ER-02.11
backup fuel	LDO LHV	42.9	MJ/kg	range 42.6-43.2	SRC-OWNR-0004, ER-03.07
backup fuel	LDO sulphur	10	mg/kg	maximum	SRC-OWNR-0004, ER-03.07
backup fuel	LDO Na + K at GT inlet	0.5	mg/kg	maximum; as delivered up to 1.0 (history 1.8)	SRC-OWNR-0004, ER-03.07
backup fuel	LDO vanadium / lead / calcium at GT inlet	0.5 / 1.0 / 2.0	mg/kg	maximum	SRC-OWNR-0004, ER-03.07
backup fuel	LDO autonomy at base load	72	h	two tanks of 50 %	SRC-OWNR-0004, ER-03.08
backup fuel	LDO annual operating limit	500	h/year	back-up fuel status	SRC-OWNR-0004, SRC-OWNR-0007
backup fuel	LDO resupply capacity	25	trucks/day	30-36 m3 each	SRC-OWNR-0004
emissions	NOx (as NO2)	30	mg/Nm3	dry, 15 % O2, 50-100 % GT load	ER-14.02
emissions	CO	30	mg/Nm3	dry, 15 % O2, 50-100 % GT load	ER-14.02
emissions	NH3 slip	5	mg/Nm3	dry, 15 % O2	ER-14.02
emissions	Stack height	65	m	EIA decision (approved dispersion study)	ER-06.06, SRC-OWNR-0007
emissions	NOx on LDO	50	mg/Nm3	dry, 15 % O2	SRC-OWNR-0007, ER-14.02
emissions	SO2 on LDO	10	mg/Nm3	dry, 15 % O2	SRC-OWNR-0007, ER-14.02
emissions	Dust on LDO	5	mg/Nm3	dry, 15 % O2	SRC-OWNR-0007, ER-14.02
fuel gas	Reference fuel LHV	48.4	MJ/kg	reference composition	ER-03.02
fuel gas	Fuel LHV range	44.2 to 50.0	MJ/kg		ER-03.02
fuel gas	Wobbe index range	47 to 52	MJ/Nm3		ER-03.02
fuel gas	Hydrogen admixture capability	5	vol %	without modification; 10 vol % by future OEM retrofit (provisions included)	ER-03.02
fuel gas	Gas supply pressure at terminal point	45 to 70	barg		ER-03.03
fuel gas	Gas supply temperature at terminal point	5 to 25	degC		ER-03.03
fuel gas	Methane content range	85.0 to 98.5	mol %		SRC-OWNR-0003, ER-03.02
fuel gas	Total sulphur limit	30	mg S/Sm3	measured typical 4, max 17	SRC-OWNR-0003, ER-03.06
fuel gas	H2S + COS limit	5	mg/Sm3		SRC-OWNR-0003, ER-03.06
fuel gas	Mercury limit	1	ug/Sm3	measured < 0.01	SRC-OWNR-0003, ER-03.06
fuel gas	Na + K, Pb, V in gas (network data)	< 0.005 (detection limit)	mg/Sm3	not guaranteed by network operator	SRC-OWNR-0003, ER-03.06
fuel gas	Fuel gas superheat above dew points at GT skid	28	K	minimum	ER-03.06
fuel gas	Water dew point	-8	degC at 70	maximum	SRC-OWNR-0003

Category	Parameter	Value	Unit	Condition	Basis
			barg		
fuel gas	Hydrocarbon dew point	-2	degC (1-70 barg)	maximum	SRC-OWNR-0003
fuel gas	Exceptional minimum supply pressure	40	barg	max 72 h/year, 24 h notice	SRC-OWNR-0003
fuel gas	Booked gas capacity (firm)	115000	Sm3/h	130,000 interruptible	SRC-OWNR-0003
grid	Grid connection voltage	380	kV	nominal	ER-11.01
grid	Switchyard short-circuit rating	50	kA		ER-11.01
grid	Grid frequency	50	Hz	operating range 47.5 to 52.5 Hz per Grid Code	ER-04.07
grid	Generator power factor range	0.85 lagging to 0.95 leading	-	at rated active power	ER-04.08
grid	Short-circuit level at 380 kV (2029)	12 (min) to 40 (max)	kA	future 50 kA (2035)	SRC-OWNR-0006
grid	380 kV voltage range	360 to 410	kV	342-420 kV for up to 30 min	SRC-OWNR-0006
grid	Connection capacity	640	MW export	40 MW import	SRC-OWNR-0006
noise	Near-field noise	85	dB(A)	1 m from equipment	ER-14.03
noise	Noise at nearest receptor (night)	45	dB(A)	LAeq night	ER-14.04
operation	Design life	30	years		ER-01.03
operation	Operating hours per year	8000	EOH/year		ER-01.03
operation	Starts per year	250	starts/year	50 cold, 100 warm, 100 hot	ER-01.03
operation	Hot start time to base load	45	min	maximum	ER-04.04
operation	Warm start time to base load	90	min	maximum	ER-04.04
operation	Cold start time to base load	180	min	maximum	ER-04.04
performance	Net electrical output (guaranteed)	600	MW	SRC, 380 kV terminal point	ER-04.01
performance	Net heat rate LHV (guaranteed)	5860	kJ/kWh	SRC	ER-04.02
performance	Net heat rate LHV at 60 % load (guaranteed)	6250	kJ/kWh	SRC	ER-04.03
performance	Auxiliary power consumption	2.2	% of gross	maximum, base load	ER-04.10
performance	Load ramp rate	35	MW/min	minimum	ER-04.05
performance	Minimum environmental load	40	% net output	maximum	ER-04.06
site	Site grade level	15	m a.s.l.		ER-02.01
site	Seismic design PGA (DD-2, 475-year)	0.45	g	TBDY 2018, site-specific hazard study	ER-02.06
site	Atmospheric corrosivity category	C5 (ISO 12944-2), durability H	-		ER-02.08
site	Basic wind speed	30	m/s	10 min mean, 10 m, 50-year return	ER-02.05
site	Ground snow load	0.75	kN/m2		ER-02.05
site	Design rainfall intensity	110	mm/h	10 min, 25-year return	ER-02.05
site	Plot dimensions	400 x 300	m	12.0 ha; usable about 10.6 ha	SRC-OWNR-0005, ER-02.01
site	Heavy transport road limit	120	t gross	two bridges on the state road	SRC-OWNR-0005, ER-02.14
site	Distance to heavy-lift port	18	km by sea	400 t quay crane	SRC-OWNR-0005
site	Groundwater level	13.5	m a.s.l.	brackish, chloride 3,000-8,000 mg/l	SRC-OWNR-0008
site	Seismic site class	ZD (Vs30 = 240 m/s)	-	TBDY 2018	SRC-OWNR-0008
water	Seawater temperature range at intake	6 to 27	degC		ER-02.04
water	SRC seawater temperature	16	degC	Site Reference Conditions	ER-02.03
water	Condenser cooling water temperature rise	7	K	maximum, base load	ER-07.04
water	Thermal plume excess at 100 m mixing zone	3	K	maximum	ER-08.05
water	Residual chlorine at outfall	0.1	mg/l	maximum	ER-08.04
water	Seawater temperature at intake - annual mean	15.4	degC	12 m depth, station M2	SRC-OWNR-0002

Category	Parameter	Value	Unit	Condition	Basis
	(measured)				
water	Seawater temperature at intake - maximum	27	degC	August	SRC-OWNR-0002
water	Seawater temperature at intake - minimum	6.9	degC	February	SRC-OWNR-0002
water	Seawater chloride - maximum (upper layer)	14500	mg/l		SRC-OWNR-0002
water	Seawater salinity range (upper layer)	21.8 to 26.1	PSU	lower layer 38.2 PSU below pycnocline	SRC-OWNR-0002
water	Pycnocline depth	20 to 25	m		SRC-OWNR-0002
water	Seawater TSS maximum (storm)	140	mg/l		SRC-OWNR-0002
water	Seawater boron	2.4 to 3.4	mg/l		SRC-OWNR-0002
water	Jellyfish load design	20	kg/m3	blooms June-September	SRC-OWNR-0002, ER-08.07
water	Wave height Hs 50-year	4.4	m	Tp 8.5 s, SW	SRC-OWNR-0002
water	Condenser pressure at SRC (maximum)	36	mbar(a)	base load, seawater 16.0 degC	ER-07.08
water	Condenser cleanliness factor (design)	0.85	-	HEI	ER-07.07
water	Condenser tube velocity	1.8 to 2.2	m/s	titanium	ER-07.07

Appendix F - Parties and Addresses for Notices

Code	Name	Role	Country	Representative	Address	Fictional
IEC	Imaginary Electric Company	consortium member	Germany	Power Island Package Manager	Fictional power-equipment manufacturer	yes
IEPC	Istanbul EPC Mühendislik ve Taahhüt A.Ş.	consortium leader	Türkiye	Consortium Project Director	Istanbul, Türkiye	yes
OE	Owner's Engineer (to be appointed)	owners engineer	Türkiye			yes
OWNER	Alpillia Enerji Üretim A.Ş.	owner	Türkiye	Project Director (Owner)	Alpillia Power Plant Site Office, Marmara coastal industrial zone, Türkiye	yes
TSO	National Transmission System Operator	tso	Türkiye			no

Appendix G - Key Dates

ID	Key date	Date	Obligation
KD-000	Notice to Proceed	2026-11-02	start no earlier than
KD-010	Back-feed energisation (key date)	2028-08-02	complete no later than
KD-020	GT first fire (key date)	2028-10-02	complete no later than
KD-030	First synchronisation in combined cycle (key date)	2029-01-02	complete no later than
KD-040	Taking-Over of the Works (Time for Completion, 975 days)	2029-07-04	complete no later than

Key dates are contractual; the Contractor's Level 3 programme (Sub-Clause 9.3) shall be linked to them.

Appendix H - Owner-provided Documents

Document	Title	Rev	Date	File in the project repository
ALP-OWN-DWG-001	External connections - gas, grid, road, water (drawing)	0	2026-09-27	sources/owner/ALP-OWN-DWG-001_Rev0.pdf
ALP-OWN-DWG-002	Site and construction area plan (drawing)	0	2026-09-27	sources/owner/ALP-OWN-DWG-002_Rev0.pdf
ALP-OWN-ELE-001	Grid connection data	0	2026-09-11	sources/owner/ALP-OWN-ELE-001_Rev0.pdf
ALP-OWN-ENV-001	Positive EIA decision - binding conditions	0	2026-09-09	sources/owner/ALP-OWN-ENV-001_Rev0.pdf
ALP-OWN-FUL-001	Natural gas quality and supply conditions	0	2026-09-08	sources/owner/ALP-OWN-FUL-001_Rev0.pdf
ALP-OWN-FUL-002	Secondary fuel: light diesel oil (LDO)	0	2026-09-15	sources/owner/ALP-OWN-FUL-002_Rev0.pdf
ALP-OWN-GEN-001	Site, existing infrastructure and tie-in points	1	2026-09-27	sources/owner/ALP-OWN-GEN-001_Rev1.pdf
ALP-OWN-GEO-001	Geotechnical baseline summary	0	2026-09-05	sources/owner/ALP-OWN-GEO-001_Rev0.pdf
ALP-OWN-MAR-001	Seawater quality and marine conditions	0	2026-09-12	sources/owner/ALP-OWN-MAR-001_Rev0.pdf
ALP-OWN-MET-001	Meteorological and ambient air data	0	2026-09-10	sources/owner/ALP-OWN-MET-001_Rev0.pdf

The Owner's documents are part of the Contract to the extent stated in Article A3; the Contractor is responsible for their interpretation (Sub-Clause 4.5).

Appendix I - Agreed Clarifications

Technical queries raised by the Contractor and answered by the Owner before signature. Where an answer modifies Appendix A or an Owner document, the answer prevails (Article A3). Records changed as a result are listed; the changed requirements are already reflected in Appendix A.

TQ	Round	Subject	Question	Owner's answer	Outcome	Price	Records changed
TQ-001	1	Site Reference seawater temperature 16.0 degC vs measured 15.4 degC	ER-02.03 and ER-07.08 fix the Site Reference seawater temperature at 16.0 degC. ALP-OWN-MAR-001 section 3 gives a measured annual mean of 15.4 degC at the intake depth. Which value is the contractual reference for PG-01/PG-02 and the condenser design?	Confirmed: 16.0 degC remains the contractual Site Reference seawater temperature for all guarantees. The measured data in ALP-OWN-MAR-001 (annual mean 15.4 degC, range 6.9-27.0 degC) is the design range for the condenser and CW system. ER-02.03 is annotated accordingly.	confirmed as is	-	requirement/ER-02.03
TQ-002	1	Gas LHV range: 45.0 MJ/kg (contract) vs 44.2 MJ/kg (network data)	ER-03.02 requires operation for LHV 45.0-50.0 MJ/kg; ALP-OWN-FUL-001 shows 44.2 MJ/kg for 3 % of hours (high N2/CO2 gas). Must the plant operate, and meet guarantees and emissions, below 45.0 MJ/kg?	Accepted. The plant shall operate and meet the emission limits for LHV 44.2-50.0 MJ/kg; performance guarantees apply at the reference gas with corrections per the agreed curves.	requirement changed	-	requirement/ER-03.02, design_parameter/D P-CASE-0019
TQ-003	1	Exceptional minimum gas pressure 40 barg is below the gas turbine requirement	The GT requires at least 43.5 barg at the fuel skid inlet at base load; with about 1.5 bar loss in the gas receiving station we need 45 barg at TP-G1. ALP-OWN-FUL-001 allows 40 barg for up to 72 h/year. Is a gas booster compressor expected, or may the plant transfer to LDO / reduce load during these periods?	Superseded by TQ-017: load limitation between 40 and 45 barg, no compressor.	contractor proposal accepted	-	
TQ-004	1	Hydrogen admixture up to 10 vol %	The offered DLN combustion system is released for 5 vol % hydrogen. 10 vol % requires an upgraded combustor set that will be available as a retrofit. ALP-OWN-FUL-001 says 10 % is not firm	Accepted: 5 vol % hydrogen without modification and with all guarantees; 10 vol % by future OEM retrofit (not in the Contract Price) with provisions now as proposed.	contractor proposal accepted	-	requirement/ER-03.02, design_parameter/D P-CASE-0021

TQ	Round	Subject	Question	Owner's answer	Outcome	Price	Records changed
			before 2035. Please confirm the requirement.				
TQ-005	1	LDO alkali metals: delivered 1.0 mg/kg (history 1.8) vs 0.5 mg/kg at GT inlet	ER-03.07 requires Na+K <= 0.5 mg/kg at the GT inlet; ALP-OWN-FUL-002 says delivered LDO may contain up to 1.0 mg/kg and has reached 1.8 mg/kg. ER-03.09 asks for treatment 'if needed'. Is the Owner's intent that treatment is always included?	Rejected in round 1; resolved by the compromise in TQ-018.	price adjustment	-	
TQ-006	1	LDO storage volume and resupply rate	72 h at base load on LDO is about 6,900 m3 (two tanks of about 3,500 m3 net). The supplier's maximum of 25 trucks/day (about 825 m3/day) covers only about 36 % of base-load consumption. Please confirm that the plant is not required to run beyond the storage autonomy, and that 'net working volume' excludes heel and fire-fighting reserves.	Confirmed: two tanks of 3,500 m3 net working volume each (heel and fire-fighting reserves in addition); the plant is not required to operate beyond the storage autonomy.	requirement changed	-	requirement/ER-03.08
TQ-007	1	Stack height 60 m (contract) vs 65 m (EIA decision)	ER-06.06 and Appendix E state 60 m; ALP-OWN-ENV-001 requires 65 m based on the approved dispersion study. Please confirm 65 m.	Accepted: stack height 65 m as required by the EIA decision; price adjustment EUR 180,000 accepted.	price adjustment	EUR 180,000	requirement/ER-06.06, design_parameter/D P-CASE-0043
TQ-008	1	Sanitary wastewater: no municipal sewer at the Site	SC-043 assigns 'potable water and sewage connections at site boundary' to the Owner, but ALP-OWN-GEN-001 states that no sewer is available (TP-S1). Who provides sanitary wastewater treatment and discharge?	Accepted: the Owner provides potable water only. The Contractor provides a packaged sanitary wastewater treatment plant with discharge through the outfall within the EIA limits; EUR 350,000 accepted.	price adjustment	EUR 350,000	scope_item/SC-043, scope_item/SC-057, tie_in/TP-S1, requirement/ER-09.05
TQ-009	1	Heavy transport: no jetty, 120 t bridges	The GT (about 460 t), generators (about 330 t) and GSU transformers (about 280 t) cannot use the state road. We plan a temporary barge landing on the 60 m public coastal strip. ALP-OWN-GEN-001 says the Owner can apply for the coastal permit (about 6 months). Please confirm the Owner obtains the permit and the dates.	Confirmed: the Contractor designs, builds, operates and removes the temporary barge landing at its cost (ER-02.14). The Owner applies for the coastal permit within 30 days of receiving the Contractor's permit design and will use reasonable endeavours to obtain it by 2027-09-30. Please state when the permit design will be submitted.	owner to provide	-	requirement/ER-02.14, tie_in/TP-M1
TQ-010	1	380 kV line readiness 2028-09-30 is after the back-feed key date 2028-08-02	Key date KD-010 (back-feed, 2028-08-02) and first fire KD-020 (2028-10-02) need the 380 kV line, which ALP-OWN-ELE-001 and TP-E1 give as ready on 2028-09-30. Please advance the line or move the key dates.	The Owner will accelerate its 380 kV line contract: TP-E1 will be available by 2028-07-15. The key dates remain unchanged.	owner to provide	-	tie_in/TP-E1
TQ-011	1	Marine works window and marine permit date	Marine works are prohibited 1 May - 30 September (ALP-OWN-ENV-001). With the marine permit by 2027-04-30 (TP-CW1/CW2) the first usable window is October 2027 - April 2028, which must also carry the barge landing. Can the permit be obtained by 2027-02-28 to use part of the spring 2027 window?	The contractual marine permit date remains 2027-04-30. The Owner will target 2027-02-28 without commitment.	confirmed as is	-	
TQ-012	1	Construction laydown area	The usable plot area is about 10.6 ha after setbacks, which is not enough for the power block, laydown, pre-assembly and contractor facilities. Will the Owner exercise its option on the 5 ha field north of the plot?	Accepted: the Owner will exercise the option; the 5 ha field north of the plot is available free of charge from 2027-03-01 until Taking-Over; the Contractor restores it to its original condition.	owner to provide	-	tie_in/TP-A1, requirement/ER-02.01

TQ	Round	Subject	Question	Owner's answer	Outcome	Price	Records changed
TQ-013	1	Verification of salt removal at the GT air inlet	ER-02.10 requires Na+K at the compressor inlet within the OEM limit at the storm maximum (165 ug/m3). This cannot be tested on demand. Please accept verification by analysis plus monitoring.	Accepted as proposed: verification by filter design report with OEM confirmation plus 12 months of Na+K air sampling downstream of the filters, reported during the Defects Notification Period.	contractor proposal accepted	-	requirement/ER-02.10
TQ-014	1	Status of the 36 mbar(a) condenser pressure	ER-07.08 sets 36 mbar(a) at SRC and names an ASME PTC 12.2 test, but there is no guarantee or LD for it. Is the condenser pressure a separate guarantee, or is compliance demonstrated through the plant heat rate guarantee?	Accepted: the condenser pressure is a design requirement; contractual compliance is through PG-02. The ASME PTC 12.2 test is carried out for information.	contractor proposal accepted	-	requirement/ER-07.08
TQ-015	1	380 kV switchgear in a C5 salt atmosphere	ER-11.01 asks for a GIS switchyard and ER-02.10 for very-heavy-pollution creepage outdoors. Is an indoor GIS in a building acceptable to the Owner?	Accepted: indoor GIS in a ventilated building, only the line gantry bushings outdoors; no price change.	contractor proposal accepted	-	requirement/ER-11.01
TQ-016	1	Relocation of the buried 11 kV cable	The 11 kV cable along the east boundary is relocated only by 2027-06-30 (ALP-OWN-GEN-001). Our layout places the LDO tank farm and warehouse in the east. Can the relocation be completed by 2027-02-28?	Not possible: the relocation is carried out by the regional utility and 2027-06-30 is its committed date. The Contractor shall sequence the east-area works accordingly.	rejected	-	
TQ-017	2	Low gas pressure: booster compressor option vs load limitation (follows TQ-003)	As requested: (a) booster compressor option, 2 x 100 % electric reciprocating units, 115,000 Sm3/h from 40 to 47 barg, about 1.3 MW each: EUR 6,800,000, 14 months delivery, no key-date impact if ordered by 2027-03-31. (b) Estimated operation below 45 barg: about 40 h/year based on the network data. (c) Zero-cost alternative: between 40 and 45 barg the GT reduces load to the pressure-limited load (about 60 % GT load at 40 barg) and stays on gas; transfer to LDO only below 40 barg. Which does the Owner choose?	The Owner chooses alternative (c): no booster compressor. Between 40 and 45 barg the plant operates on gas at the pressure-limited load; automatic transfer to LDO only below 40 barg. ER-03.03 is revised. TQ-003 is closed by this answer.	contractor proposal accepted	-	requirement/ER-03.03, clarification/TQ-003
TQ-018	2	LDO treatment - request to reconsider (follows TQ-005)	ALP-OWN-FUL-002 lists Na+K <= 1.0 mg/kg 'as delivered, not guaranteed by supplier' in the specification table; our tender assumed the tabulated value, which the forwarding filters and water draining handle for most deliveries. The 1.8 mg/kg history is a note on other plants. The requirement 'if needed' is ambiguous for a lump-sum price. We propose a compromise.	Accepted as a compromise: fuel treatment (2 x 100 % centrifugal separators with water washing) becomes mandatory scope; cost shared, price adjustment EUR 950,000. The Owner will require Na+K <= 1.0 mg/kg with sampling of each delivery in its LDO supply contract (decision recorded). TQ-005 is closed by this answer.	price adjustment	EUR 950,000	requirement/ER-03.09, scope_item/SC-053, clarification/TQ-005, decision (LDO supply contract)
TQ-019	2	Barge landing permit: submission and application dates (follows TQ-009)	We will submit the barge landing permit design by 2027-03-31. Please confirm the Owner's application by 2027-04-30 and that the crossing of the public coastal strip by the intake and outfall pipelines is covered by the marine permit.	Confirmed: the Owner applies for the coastal permit by 2027-04-30 provided the design is received by 2027-03-31; the crossing of the coastal strip by the intake and outfall pipelines is covered by the marine permit. TQ-009 is closed.	owner to provide	-	tie_in/TP-M1, clarification/TQ-009
TQ-020	2	Contractual status of the 380 kV line date 2028-07-15 (follows TQ-010)	Please confirm that TP-E1 availability by 2028-07-15 is an Owner obligation under Sub-Clause 2.4 (so that a late line entitles the Contractor to an extension of time under Sub-Clause 9.4), and who coordinates the line energisation with the TSO.	Confirmed: availability of TP-E1 by 2028-07-15 is an Owner-furnished item under Sub-Clause 2.4; late availability entitles the Contractor to relief under Sub-Clause 9.4. The Owner energises the line with the TSO; the Contractor coordinates the plant-side protection settings	owner to provide	-	requirement/ER-18.02, clarification/TQ-010

TQ	Round	Subject	Question	Owner's answer	Outcome	Price	Records changed
				and tests. TQ-010 is closed.			
TQ-021	2	Demineralised water for LDO water injection	Water injection needs about 80 t/h of demineralised water at base load on LDO. ER-09.02 requires storage for 72 h of LDO operation (about 5,800 m3) on top of 3 days of cycle make-up: a very large tank for a fuel used less than 500 h/year.	Accepted as proposed: demin storage 2 x 1,250 m3, demin plant sized for continuous water injection at 50 % LDO load, and a connection for mobile demin trailers; no price change.	contractor proposal accepted	-	requirement/ER-09.02
TQ-022	2	Seawater temperature correction in the performance test (follows TQ-001)	Following TQ-001, please confirm that the performance test corrects from the measured seawater temperature during the test to the 16.0 degC reference using the agreed correction curves.	Confirmed: corrections from the measured test seawater temperature to 16.0 degC with the agreed correction curves.	confirmed as is	-	
TQ-023	3	Reference LDO for the LDO performance guarantees	ALP-OWN-FUL-002 allows FAME (biodiesel) up to 7 % v/v and LHV 42.6-43.2 MJ/kg. PG-15/PG-16 refer to LHV 42.9 MJ/kg but not to FAME or water content, which affect output, heat rate and water injection. Please define the reference LDO for the guarantees.	Accepted as proposed; PG-15 and PG-16 conditions revised.	contractor proposal accepted	-	guarantee/PG-15, guarantee/PG-16
TQ-024	3	Consolidated price and incorporation of the clarifications	All technical queries TQ-001 to TQ-023 are closed or answered. Agreed price adjustments: TQ-007 EUR 180,000, TQ-008 EUR 350,000, TQ-018 EUR 950,000; total EUR 1,480,000 (onshore portion). No time impacts. We propose the Contract Price of EUR 532,480,000 and that the clarification register becomes Appendix I of the Contract, prevailing over Appendix A where it modifies it. Please confirm readiness for signature on 2026-10-15.	Confirmed. Contract Price EUR 532,480,000 (offshore EUR 327,000,000; onshore EUR 205,480,000); Appendix H (Owner-provided documents) and Appendix I (agreed clarifications) are added to the contract documents in Article A3. The Contract is agreed and ready for signature on 2026-10-15.	price adjustment	-	contract/ALP-EPC-001, contract_clause/ALP-EPC-001_A3, contract_clause/ALP-EPC-001_A4
TQ-025	4	Reference fuel LHV inconsistent with the reference composition	ER-03.02 and DP-CASE-0018 give LHV 48.4 MJ/kg for the reference composition. ISO 6976 gives 47.48 MJ/kg (34.78 MJ/Sm3, relative density 0.598) for that composition; ALP-OWN-FUL-001 section 3 itself gives 34.4 MJ/Sm3 and relative density 0.600, which also mean about 47 MJ/kg. Guarantees are on heat input, so the error does not change them, but fuel flow, the gas booking check and the metering design use the mass LHV.	(open)	-	-	
TQ-026	4	Power island control system: integration solution for SC-013	IEC's turbine control system keeps its own controllers (turbine protection) with the HMI integration package: graphics and alarms native on the plant DCS operator stations, one KKS tag database, common time sync, own IEC 62443 zone (TQ-IEC-004). Does the Owner accept this as 'integrated in the DCS' under SC-013?	(open)	-	-	
TQ-027	4	Export above the 640 MW connection capacity in winter	Below about 5 degC the plant can export more than 640 MW (about 670 MW net at -4.8 degC). The TCS provides an export limiter at the HV metering point. Please confirm the export limit to be set (640 MW net at the 380 kV meters?) and whether the Owner intends to increase the	The export limiter shall be set to 640 MW net at the 380 kV revenue meters. The Owner has applied to the TSO for a connection capacity of 680 MW; if granted, the Owner will instruct the new set point as a variation.	contractor proposal accepted	-	tie_in/TP-E1

TQ	Round	Subject	Question	Owner's answer	Outcome	Price	Records changed
			connection capacity.				
TQ-028	4	Hot gas path interval: EOH basis and calendar interval	ER-05.07 requires a hot gas path interval of at least 25,000 EOH. With the OEM start factors (hot 10, warm 15, cold 20 EOH) and the ER-01.03 regime this is about 2.3 calendar years. Please confirm that ER-05.07 is EOH-based as written.	(open)	-	-	
TQ-029	4	Scope item SC-011 still states a 60 m stack	TQ-007 agreed a 65 m stack and ER-06.06 was revised, but the scope split item SC-011 (Appendix D) still reads "Stack 60 m with CEMS platform and silencer". Please confirm the editorial correction before signature.	(open)	-	-	
TQ-030	5	Tie-in register: confirmation of the terminal point conditions	The Consortium's tie-in register (draft) states for each terminal point the exact scope break, connection, isolation, protection, operating envelope, capacity and the dates needed from the Contractor's commissioning sequence. Please confirm, and provide the missing data: (a) TP-W1 maximum supply pressure; (b) TP-E2 prospective short-circuit current of the 34.5 kV supply; (c) SC-001 (site boundary) has no terminal point - propose TP-A2 plot handover incl. the 11 kV cable relocation by 2027-06-30; (d) SC-035 and SC-038 need Owner/TSO/authority input - propose TP-T3 (TSO SCADA and metering link) and TP-T2 (emission data link to the Ministry); (e) TP-A1 is available on 2027-03-01, the day it is needed (no float): earlier access requested.	Register confirmed with: (a) TP-W1 2.5-6.0 barg; (b) TP-E2 12.5 kA; (c) TP-A2 added (plot, 11 kV cable relocated by 2027-06-30, east strip E 370-400 until then not available); (d) TP-T2 and TP-T3 added; (e) TP-A1 lease from 2027-02-01. The conditions of all terminal points TP-A1..TP-W1 are agreed.	contractor proposal accepted	-	tie_in/TP-A1, tie_in/TP-A2, tie_in/TP-CW1, tie_in/TP-CW2, tie_in/TP-D1, tie_in/TP-E1, tie_in/TP-E2, tie_in/TP-G1, tie_in/TP-L1, tie_in/TP-M1, tie_in/TP-R1, tie_in/TP-S1, tie_in/TP-T1, tie_in/TP-T2, tie_in/TP-T3, tie_in/TP-W1
TQ-031	5	Short-circuit level at TP-E1: plant contribution	ALP-OWN-ELE-001 gives 40 kA (2029) and 50 kA (2035 planning) at the plant 380 kV busbar; the GIS is specified for 50 kA (ER-11.01). The Alpillá generators add about 3.7 kA (GT 2.4 kA, ST 1.3 kA). Please confirm whether the TSO values include the Alpillá contribution; if not, 50 kA would be exceeded in 2035.	The TSO confirms (letter of 2026-09-25) that 40 kA (2029) and 50 kA (2035) include the contribution of the Alpillá units. The GIS rating of 50 kA stands.	confirmed as is	-	tie_in/TP-E1
TQ-032	5	Reactive capability: generator terminals (ER-04.08) or connection point (ELE-001)	ER-04.08 requires 0.85 lagging / 0.95 leading at rated active power at the generator terminals; ALP-OWN-ELE-001 section 3 states the same at the connection point. At the connection point, at the winter GT output of 470 MW, the GT generator and GSU would need about 600 MVA instead of 560 MVA (GSU reactive losses). Please confirm that ER-04.08 governs.	ER-04.08 governs: 0.85 lagging / 0.95 leading at rated active power at the generator terminals. The TSO accepts the Grid Code compliance demonstration on that basis; ALP-OWN-ELE-001 will be corrected in its next revision.	confirmed as is	-	